STM32U575I-EV

MB1550

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Legend

General comment such as function title, configuration, ...

Text to be added to silkscreen.

Warning text.

Notes to generate the board layout.

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U_MB1550_TOP
MB1550_TOP.SchDoc

HW100
STICKER BOARD

HW101
STICKER PRODUCT

Title: **Project overview**			
Project: **STM32U575I-EV**			
Variant: U575AIQ			
Revision: C-02		Reference: MB1550	
Size: A4	Date: 02 Sept 2021	Sheet: 1 of 24	

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U_MFX_IO_expander
MFX_IO_expander.SchDoc
LCD_RESET
MFX_WAKEUP
MFX_IRQ_OUT
NRST
I2C_SCL
I2C_SDA
SMARTCARD_3/5V
SMARTCARD_OFF
SMARTCARD_RST
SMARTCARD_CMDVCC
CODEC_RSTN
MEMS_LED
MFX_TSC_XP
MFX_TSC_XN
MFX_TSC_YP
MFX_TSC_YN
STMOD+_RST
EXT_RESET
U_SENSOR and I2C-interface
SENSOR and I2C-interface.SchDoc
EXT_RESET
GYRO_ACC_INT
I2C_SCL
I2C_SDA
I2C_SCL_SENSOR
I2C_SDA_SENSOR
U_OCTOSPI
OCTOSPI.SchDoc
OCTOSPI_1_IO[0..7]
OCTOSPI_1_NCS
OCTOSPI_1_DQS
OCTOSPI_1_CLK_P
OCTOSPI_1_CLK_N
OCTOSPI_2_IO[0..7]
OCTOSPI_2_NCS
OCTOSPI_2_DQS
OCTOSPI_2_CLK
NRST
U_SMARTCARD
SMARTCARD.SchDoc
SmartCard_3/5V
SMARTCARD_OFF
SMARTCARD_RST
SMARTCARD_CMDVCC
SMARTCARD_IO
SMARTCARD_CLK
U_AUDIO
AUDIO.SchDoc
CODEC_RSTN
MEMS_LED
I2C_SCL
I2C_SDA
ADF1_SDIN0
ADF1_CCK0
SAI1_SDOUT_A
SAI1_SDIN_B
SAI1_MCLK_B
SAI1_SCK_B
SAI1_FS_B
MDF1_CCK0
MDF1_CCK1
MDF1_SDIN0
MDF1_SDIN3
MDF1_SDIN5
U_STPMS2
STPMS2.SchDoc
MDF1_CCK1
MDF1_SDIN3
MDF1_SDIN5
U_PMOD_STMOD+
PMOD_STMOD+.SchDoc
MDF1_CCK1
MDF1_SDIN3
MDF1_SDIN5
SPI_NSS
SPI3_SCK
SPI3_MOSI
SPI3_MISO
SPI3_MOSI2
SPI3_MISO2
I2C_SCL
I2C_SDA
STMOD+_RST
STMOD+_INT
STMOD+_PWM
STMOD+_ADC
UART_TX
UART_RX
UART_RTS
UART_CTS
U_FDCAN_RS232
FDCAN_RS232.SchDoc
UART_TX
UART_RX
UART_RTS
UART_CTS
FDCAN_TX
FDCAN_RX
BOOTLOADER_RST
BOOTLOADER_BOOT0
U_SDCARD
SDCARD.SchDoc
SDCARD_DETECT
SDIO1_D[0..3]
SDIO1_CLK
SDIO1_CMD
U_PERIPHERAL
PERIPHERAL.SchDoc
LED_BLUE
LED_YELLOW
JOY_SEL
JOY_DOWN
JOY_LEFT
JOY_RIGHT
JOY_UP
EXT_RESET
LED_GREEN
LED_RED
TKEY
SHIELD
SHIELD_CS
TKEY_CS
ADC_DAC
COMP2_INP
COMP2_OUT
OPAMP1_INP
OPAMP1_INM
OPAMP1_VOUT
U_Camera
Camera.SchDoc
I2C_SCL
I2C_SDA
DCMI_D[0..7]
DCMI_PIXCLK
DCMI_VSYNC
DCMI_HSYNC
CAMERA_PLUG
CAMERA_RSTI
CAMERA_XSDN
I2C_SCL_SENSOR
I2C_SDA_SENSOR
U_MCU_IO
MCU_IO.SchDoc
MFX_IRQ_OUT
MFX_WAKEUP
I2C_SCL
I2C_SDA
NRST
OCTOSPI_1_IO[0..7]
OCTOSPI_1_NCS
OCTOSPI_1_DQS
OCTOSPI_1_CLK_P
OCTOSPI_1_CLK_N
OCTOSPI_2_IO[0..7]
OCTOSPI_2_NCS
OCTOSPI_2_DQS
OCTOSPI_2_CLK
SMARTCARD_IO
SMARTCARD_CLK
SAI1_SDOUT_A
SAI1_SDIN_B
SAI1_MCLK_B
SAI1_SCK_B
SAI1_FS_B
MDF1_CCK0
MDF1_CCK1
MDF1_SDIN0
MDF1_SDIN3
MDF1_SDIN5
ADF1_CCK0
ADF1_SDIN0
SPI_NSS
SPI3_SCK
SPI3_MOSI
SPI3_MISO
SPI3_MOSI2
SPI3_MISO2
STMOD+_INT
STMOD+_PWM
STMOD+_ADC
UART_TX
UART_RX
UART_RTS
UART_CTS
FDCAN_TX
FDCAN_RX
SDCARD_DETECT
SDIO1_D[0..3]
SDIO1_CLK
SDIO1_CMD
LED_GREEN
LED_RED
TKEY
SHIELD
SHIELD_CS
TKEY_CS
ADC_DAC
COMP2_INP
COMP2_OUT
OPAMP1_INP
OPAMP1_INM
OPAMP1_VOUT
GYRO_ACC_INT
I2C_SCL_SENSOR
I2C_SDA_SENSOR
DCMI_D[0..7]
DCMI_PIXCLK
DCMI_VSYNC
DCMI_HSYNC
CAMERA_PLUG
CAMERA_RSTI
CAMERA_XSDN
FMC_NE4_LCD_CSn
FMC_A22_LCD_RS
LCD_CTP_INT
LCD_BL_CTRL
LCD_TE
FMC_NOE
FMC_NWE
FMC_NE1
FMC_NBL0
FMC_NBL1
D[0..15]
A[0..20]
PA[0..15]
PB[0..15]
PC[0..15]
PD[0..15]
PE[0..15]
PF[0..15]
PG[0..15]
PH[0..15]
PI[0..7]
UCPD_PWR
USB_FS_N
USB_FS_P
UCPD_CC1
UCPD_CC2
UCPD_DBn
UCPD_FLT
UCPD_SOURCE_EN
UCPD_DISCHARGE
VBUS_SENSE
UCPD_FRSCC1
UCPD_FRSCC2
IBUS_SENSE
T_VCP_TX
T_VCP_RX
JTCK_SWCLK
JTMS_SWDIO
JTDO_SWO
JTRSTN
JTDI
TRACE_CLK
TRACE_D0
TRACE_D1
TRACE_D2
TRACE_D3
MC_EMERGENCY_STOP
MC_PWM_1H
MC_PWM_1L
MC_PWM_2H
MC_PWM_2L
MC_PWM_3H
MC_PWM_3L
MC_CURRENT_A
MC_CURRENT_B
MC_CURRENT_C
MC_ICL_SHUTOUT
MC_DISSIPATIVE_BRAKE
MC_PFC_SYNC
MC_PFC_PWM
MC_ENCODER_A
MC_ENCODER_B
MC_ENCODER_INDEX
MC_PFC_VAC
MC_PFC_SHUTDOWN
MC_TEMP
MC_PFC_IND_CURR
MC_BUS_VOLTAGE
PC14-OSC32_IN
PC15-OSC32_OUT
PH0-OSC_IN
PH1-OSC_OUT
PH3-BOOT0
TAMPER_KEY
BOOTLOADER_RST
BOOTLOADER_BOOT0
U_LCD_TFT_320x240
LCD_TFT_320x240.SchDoc
I2C_SCL
I2C_SDA
LCD_RESET
NRST
MFX_TSC_XP
MFX_TSC_XN
MFX_TSC_YP
MFX_TSC_YN
FMC_NE4_LCD_CSn
FMC_A22_LCD_RS
LCD_CTP_INT
LCD_BL_CTRL
LCD_TE
FMC_NOE
FMC_NWE
D[0..15]
U_SRAM
SRAM.SchDoc
D[0..15]
A[0..20]
FMC_NOE
FMC_NWE
FMC_NE1
FMC_NBL0
FMC_NBL1
U_Extension_Connector
Extension_Connector.SchDoc
PA[0..15]
PB[0..15]
PC[0..15]
PD[0..15]
PE[0..15]
PF[0..15]
PG[0..15]
PH[0..15]
PI[0..7]
NRST
U_USB_TYPEC
USB_TYPEC.SchDoc
USB_FS_N
USB_FS_P
UCPD_CC1
UCPD_CC2
UCPD_DBn
UCPD_FLT
UCPD_SOURCE_EN
UCPD_DISCHARGE
VBUS_SENSE
UCPD_PWR
UCPD_FRSCC1
UCPD_FRSCC2
I2C_SCL
I2C_SDA
IBUS_SENSE
U_ST_LINK_V3E
ST_LINK_V3E.SchDoc
T_VCP_TX
T_VCP_RX
JTCK_SWCLK
JTMS_SWDIO
JTDO_SWO
JTDI
NRST
JTRSTN
STLK_MCO
U_JTAG_TRACE
JTAG_TRACE.SchDoc
JTCK_SWCLK
JTMS_SWDIO
JTDO_SWO
JTRSTN
JTDI
TRACE_CLK
TRACE_D0
TRACE_D1
TRACE_D2
TRACE_D3
NRST
U_MOTOR_CONTROL
MOTOR_CONTROL.SchDoc
MC_EMERGENCY_STOP
MC_PWM_1H
MC_PWM_1L
MC_PWM_2H
MC_PWM_2L
MC_PWM_3H
MC_PWM_3L
MC_CURRENT_A
MC_CURRENT_B
MC_CURRENT_C
MC_ICL_SHUTOUT
MC_DISSIPATIVE_BRAKE
MC_PFC_SYNC
MC_PFC_PWM
MC_ENCODER_A
MC_ENCODER_B
MC_ENCODER_INDEX
MC_PFC_VAC
MC_PFC_SHUTDOWN
MC_TEMP
MC_PFC_IND_CURR
MC_BUS_VOLTAGE
U_MCU_CLOCK_&_RESET
MCU_CLOCK_&_RESET.SchDoc
PC14-OSC32_IN
PC15-OSC32_OUT
PH0-OSC_IN
PH1-OSC_OUT
PH3-BOOT0
TAMPER_KEY
BOOTLOADER_RST
BOOTLOADER_BOOT0
PH[0..15]
PC[0..15]
STLK_MCO
NRST
H3
H4
H5
H6
U_BOARD_PWR
BOARD_PWR.SchDoc
U_MCU_PWR
MCU_PWR.SchDoc

Title: **TOP: Hierarchical view**			
Project: **STM32U575I-EV**			
Variant: U575AIQ			
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STM32 MCU I/Os

HW5

DNF

CS1627113MF

U5A — MCU_IOs (A,B,C) — STM32U575AII6Q

U5B — MCU_IOs (D,E,F,G) — STM32U575AII6Q

U5C — MCU_IOs (H,I) — STM32U575AII6Q

MCU CLK & RST

VDD

R149 2K7, R147 2K7

TP7 I2C_SDA, TP8 I2C_SCL

I2C_SCL, I2C_SDA

SB46 I2C2_SCL, SB49 I2C2_SDA

SHIELD, SHIELD_CS

UCPD_CC2, SB52

USB_FS_N, USB_FS_P

PC14-OSC32_IN, PC15-OSC32_OUT

PH0-OSC_IN, PH1-OSC_OUT

NRST

PH3-BOOT0

see page: MCU CLOCK & RESET

OPAMP1_VINM, OPAMP2_VINM

TP15 OPAMP1_VINM

TP14 OPAMP2_VINM

SDIO1_CMD, MDF1_SDIN0, FMC_NOE, FMC_NWE, SAI1_SDOUT_A, FMC_NE1

LCD/SRAM, SRAM

FMC_NBL0, FMC_NBL1, TRACE_CLK

TRACE_D2, FMC_A22_LCD_RS

SAI1_SDIN_B, SAI1_MCLK_B, SAI1_SCK_B

MFX_WAKEUP

FMC_NE4_LCD_CSn

LCD_CTP_INT

UCPD_SOURCE_EN, OCTOSPI_2_DQS, DCMI_PIXCLK, OCTOSPI_2_CLK, LED_RED

UCPD_DISCHARGE

GYRO_ACC_INT

SDCARD_DETECT

DCMI_D4, DCMI_D5, DCMI_D6, DCMI_D7, OCTOSPI_2_IO2

Serial resistors are for debug purpose, to isolate IOs on extension connector from on board feature

IO PORT MUXING TABLE **BOLD: PRIMARY FEATURES (exclusive use : SB fitted)**

"ITALIC" front: SECONDARY FEATURES (can be not fitted by default)

PORT A

PA0	SB68	*OPAMP*	OPAMP1_INP	
	SB79	*STMOD+*	STMOD+_ADC	
	SB65 DNF	*MC*	MC_ENCODER_A	
	SB72 DNF	*UCPD*	IBUS_SENSE	
PA1	SB71	**STMOD+**	STMOD+_INT	
	SB78	*TAMPER_KEY*	TAMPER_KEY	
	SB67 DNF	*MC*	MC_ENCODER_B	
PA2	R151 0R	**OCTOSPI**	OCTOSPI_1_NCS	
	SB48 DNF	*MC*	MC_ENCODER_INDEX	
PA3	R150 0R	**OCTOSPI**	OCTOSPI_1_CLK_P	
	SB53 DNF	*OPAMP*	OPAMP1_VOUT	
	SB51 DNF	*MC*	MC_TEMP	
PA4	SB18	*STMOD+*	SPI_NSS	
	SB16	*ADC_DAC*	ADC_DAC	
	SB17 DNF	*MC*	MC_BUS_VOLTAGE	
PA5	SB75	**USB**	VBUS_SENSE	
	SB82	*STMOD+*	STMOD+_PWM	
PA6	R158 0R	**OCTOSPI**	OCTOSPI_1_IO3	
	SB57 DNF	*MC*	MC_PFC_VAC	
PA7	R159 0R	**OCTOSPI**	OCTOSPI_1_IO2	
	SB61 DNF	*MC*	MC_PWM_1L	
USB_FS_N	R139 DNF 0R		PA11	
USB_FS_P	R140 DNF 0R		PA12	
PA15	SB19	**USB**	UCPD_CC1	
	SB15	*JTAG*	JTDI	

PORT B

PB0	R152 0R DNF	**OCTOSPI** *MC*	OCTOSPI_1_IO1	MC_PWM_2L
PB1	R155 0R DNF	**OCTOSPI** *MC*	OCTOSPI_1_IO0	MC_PWM_3L
PB2	R148 0R DNF	**OCTOSPI** *MC*	OCTOSPI_1_DQS	MC_DISSIPATIVE_BRAKE
PB3	SB115 SB114	**AUDIO** *JTAG*		ADF1_CCK0, JTDO_SWO
PB4	SB26 SB32 SB40 SB37	**AUDIO** **JTAG** *COMP* *STMOD+*		ADF1_SDIN0, JTRSTN, COMP2_INP, SPI3_MISO2
PB5	SB39 SB36	**COMP** *STMOD+*		COMP2_OUT, SPI3_MOSI2
	JP13	*USB_DBCC1*		
PB6	SB41 SB43 DNF	*MEMS/STPMS2/STMOD+* *MC*		MDF1_SDIN5, MC_EMERGENCY_STOP
PB8		**UCPD** *FDCAN*		UCPD_FLT, FDCAN_RX
	SB119	*EXT_MEMS*		MDF1_CCK0
PB9		**UCPD** *FDCAN*		UCPD_DBn, FDCAN_TX
PB12	SB42 SB47 DNF SB44 DNF	**SMARTCARD** OCTOSPI *MC*		SMARTCARD_CLK, OCTOSPI_1_CLK_N, MC_PFC_SHUTDOWN
SHIELD	PB13	**TOUCH-KEY**		SHIELD
	JP12	RS232 / STMOD+		UART_CTS
	SB34 DNF	PB13		
SHIELD_CS	PB14 SB23 DNF	**TOUCH-KEY** PB14		SHIELD_CS
	JP9	*USB_DBCC2*		

PORT C

PC0	R160 0R DNF	**OCTOSPI** *MC*	OCTOSPI_1_IO7	MC_CURRENT_A
PC1	R163 0R DNF	**OCTOSPI** *MC*	OCTOSPI_1_IO4	MC_CURRENT_B
PC2	R162 0R DNF	**OCTOSPI** *MC*	OCTOSPI_1_IO5	MC_CURRENT_C
PC3	R164 0R DNF	**OCTOSPI** *MC*	OCTOSPI_1_IO6	MC_PFC_IND_CURR
PC4	SB116 SB117 DNF	**SMARTCARD** *LCD*		SMARTCARD_IO, LCD_TE
TKEY	PC6 SB29 DNF SB24 DNF	**TOUCH-KEY** *MC* PC6		TKEY, MC_PWM_1H
TKEY_CS	SB30 DNF SB25 DNF	**TOUCH-KEY** *MC* PC7		TKEY_CS, MC_PWM_2H
PC8	SB84 SB80 DNF	**SDIO** *MC*	SDIO1_D0	MC_PWM_3H
PC9	SB83 SB77	**SDIO** *TRACE*	SDIO1_D1	TRACE_D0
PC10	SB85 SB81	**SDIO** *TRACE*	SDIO1_D2	TRACE_D1
PC12	SB91 SB94 SB92	**SDIO** *STMOD+* *TRACE*		SDIO1_CLK, SPI3_MOSI, TRACE_D3

PORT D

PD[0..15]

PORT E

PE4	SB105	**AUDIO**		MDF1_SDIN3
	R191 0R DNF	*SRAM*	A20	

PORT F

PF9	SB62 DNF	**AUDIO** *MC*		SAI1_FS_B, MC_PFC_SYNC
PF10	SB55 SB58 DNF	**AUDIO** *MC*		MDF1_CCK1, MC_PFC_PWM

PORT G

PG6	SB38 SB33 DNF	**RS232 / STMOD+** UCPD		UART_RTS, UCPD_FRSCC1
PG7	JP10 1 2 3	LPBAM_I2C3_SCL		I2C_SCL_SENSOR
	Shunt Fitted 1-2	**RS232 / STMOD+** LPUART1		UART_TX
	SB31 DNF	UCPD		UCPD_FRSCC2
PG8	JP11 1 2 3	LPBAM_I2C3_SDA		I2C_SDA_SENSOR
	Shunt Fitted 1-2	**RS232 / STMOD+** LPUART1		UART_RX
	SB11 DNF	UCPD		UCPD_PWR
PG9	SB108 SB107 DNF	*STMOD+* *MC*		SPI3_SCK, MC_ICL_SHUTOUT
PG10	SB103 SB102	*CAMERA* *STMOD+*		CAMERA_PLUG, SPI3_MISO

PORT H

PH8	SB96	**OCTOSPI_2** *CAMERA*	OCTOSPI_2_IO3	DCMI_HSYNC
PH9	SB95	**OCTOSPI_2** *CAMERA*	OCTOSPI_2_IO4	DCMI_D0
PH10	SB97	**OCTOSPI_2** *CAMERA*	OCTOSPI_2_IO5	DCMI_D1
PH11	SB99	**OCTOSPI_2** *CAMERA*	OCTOSPI_2_IO6	DCMI_D2
PH12	SB100	**OCTOSPI_2** *CAMERA*	OCTOSPI_2_IO7	DCMI_D3

PORT I

PI2	SB101	**OCTOSPI_2** *CAMERA*	OCTOSPI_2_IO1	CAMERA_RSTI
PI3	SB98	**OCTOSPI_2** *CAMERA*	OCTOSPI_2_IO0	CAMERA_XSDN
PI5	SB112	**OCTOSPI_2** *CAMERA*	OCTOSPI_2_NCS	DCMI_VSYNC

HW7 HW8

BUSSES

- LCD/SRAM BUS: D[0..15], A[0..20]
- SDCARD BUS: SDIO1_D[0..3]
- OCTOSPI BUS: OCTOSPI_1_IO[0..7], OCTOSPI_2_IO[0..7]
- CAMERA BUS: DCMI_D[0..7]

INT

INT0 / PI0 / SDCARD_DETECT
INT1 / PA1 / STMOD+_INT
INT5 / PC5 / MFX-IRQ_OUT
INT8 / PB8 / UCPD_FLT
INT13 / PC13 / WAKEUP_KEY (User button)
INT15 / PH15 / GYRO_ACC_INT

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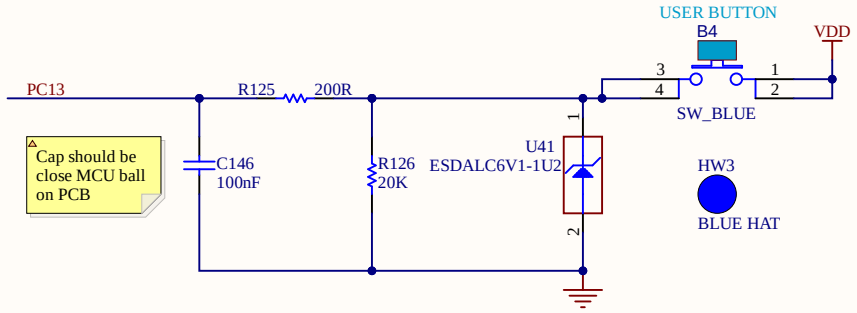
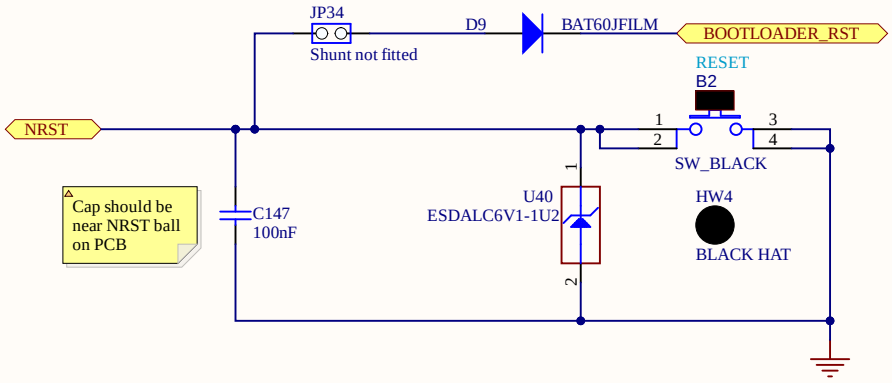
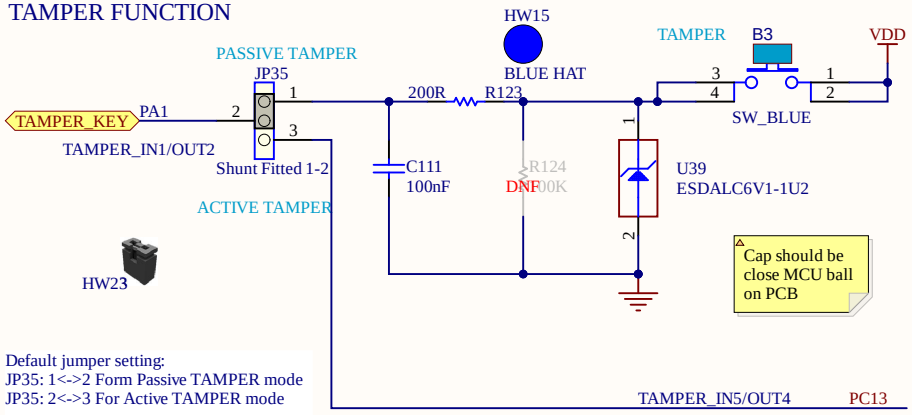
USER BUTTON



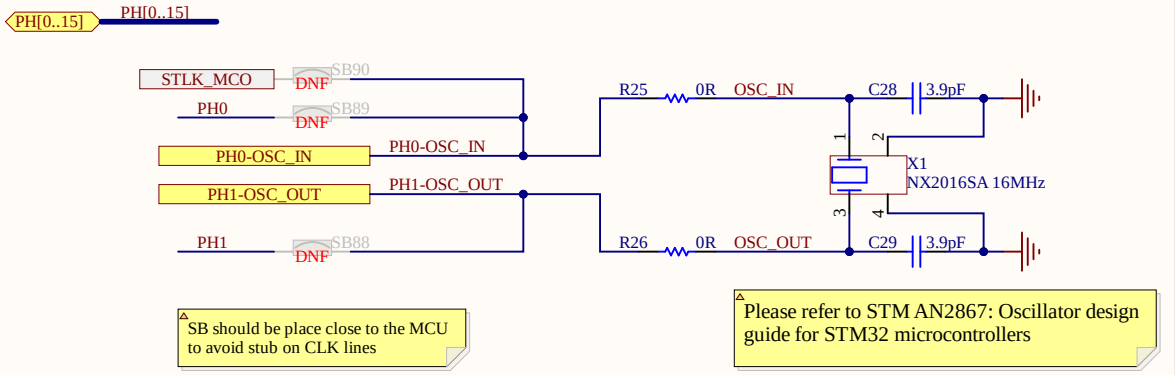
RESET FUNCTION



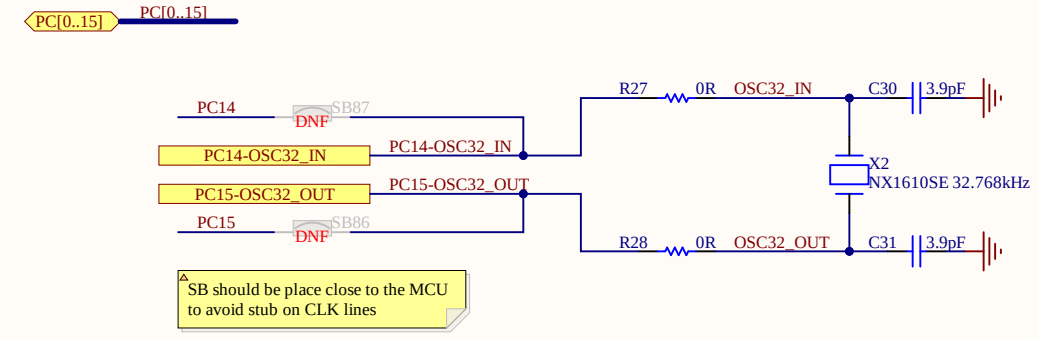
TAMPER FUNCTION



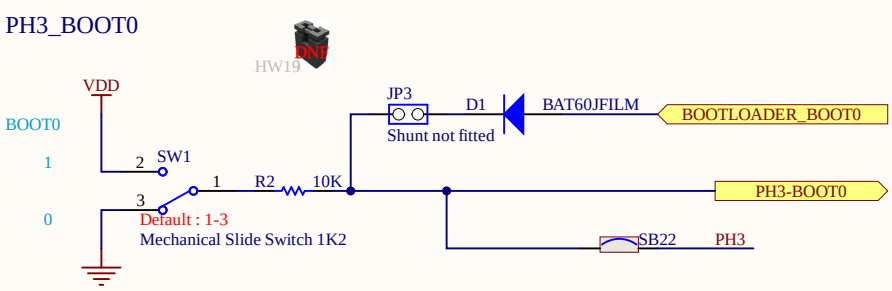
EXTERNAL HSE CLK



EXTERNAL LSE CLK



PH3_BOOT0



LIMITATIONS : See User Manual for more detail about muxing.

TAMPER on PA1 exclusive with PMOD/STMOD+ INT

Title: **MCU CLOCK & RESET**		
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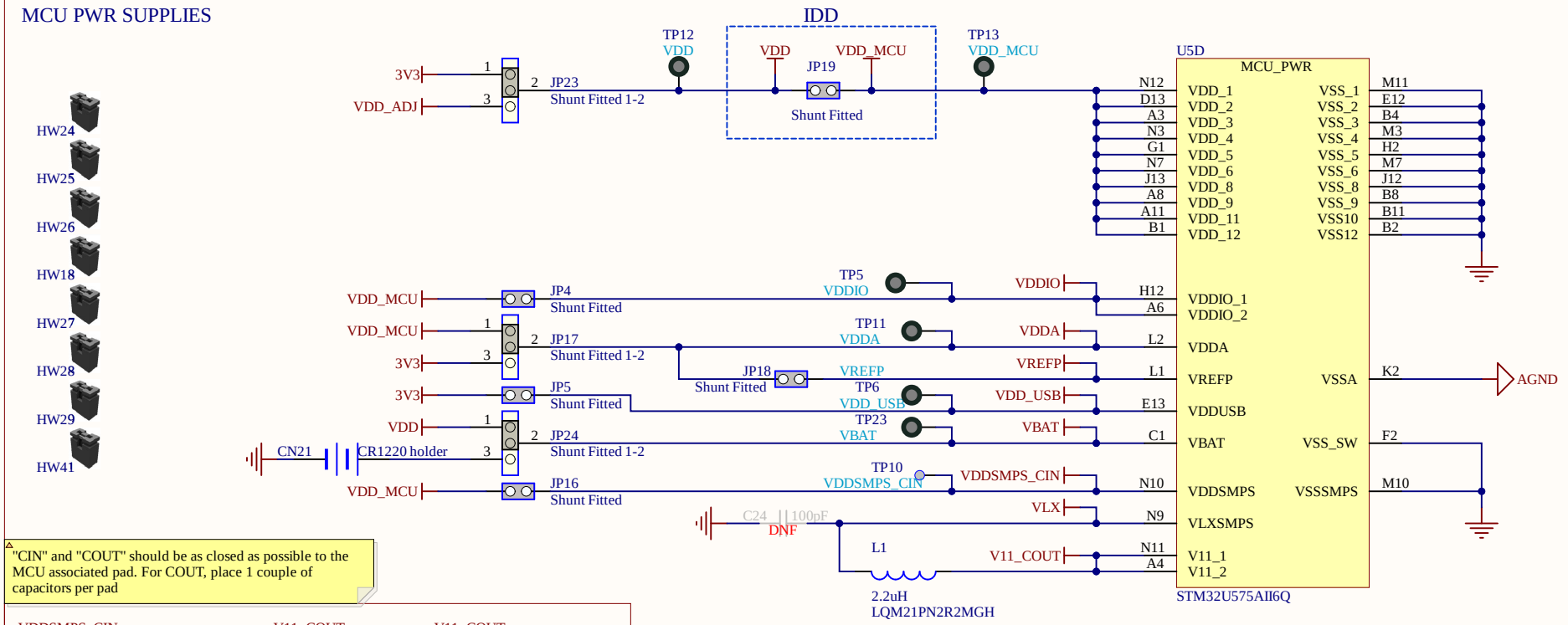
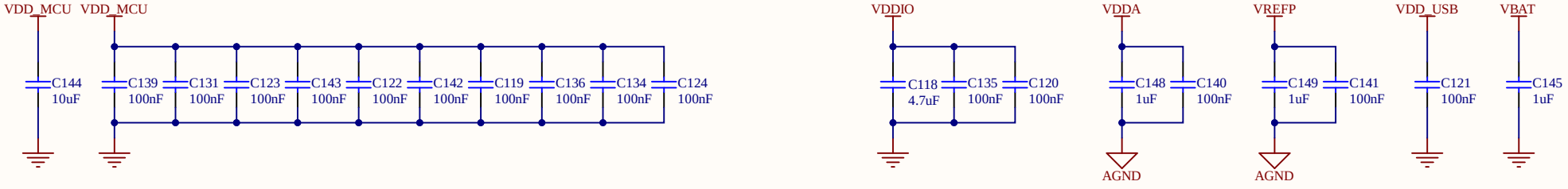
life.augmented

MCU PWR SUPPLIES



MCU DECAPS

Ceramic capacitor (Low ESR, ESR<1ohm)



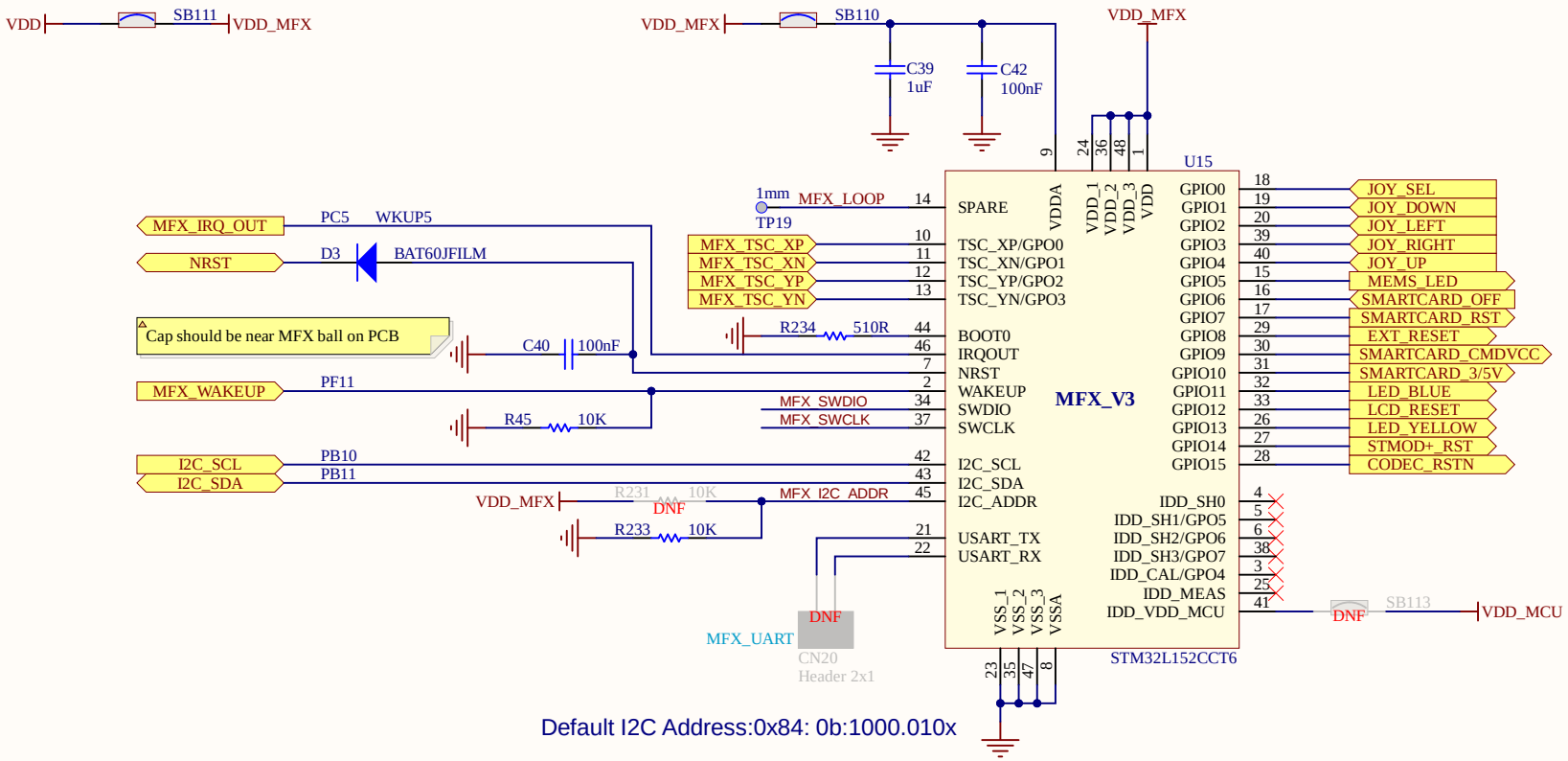
SB76 — AGND

Title: **MCU PWR**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 5 of 24

life.augmented

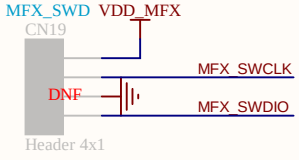
MFX I/O EXPANDER

Operating range: 1.65<VDD<3.6V

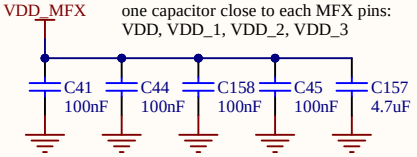


Default I2C Address:0x84: 0b:1000.010x

MFX DEBUG SWD INTERFACE



MFX DECAPS



LIMITATIONS : See User Manual for more detail about muxing.

NO LIMITATION

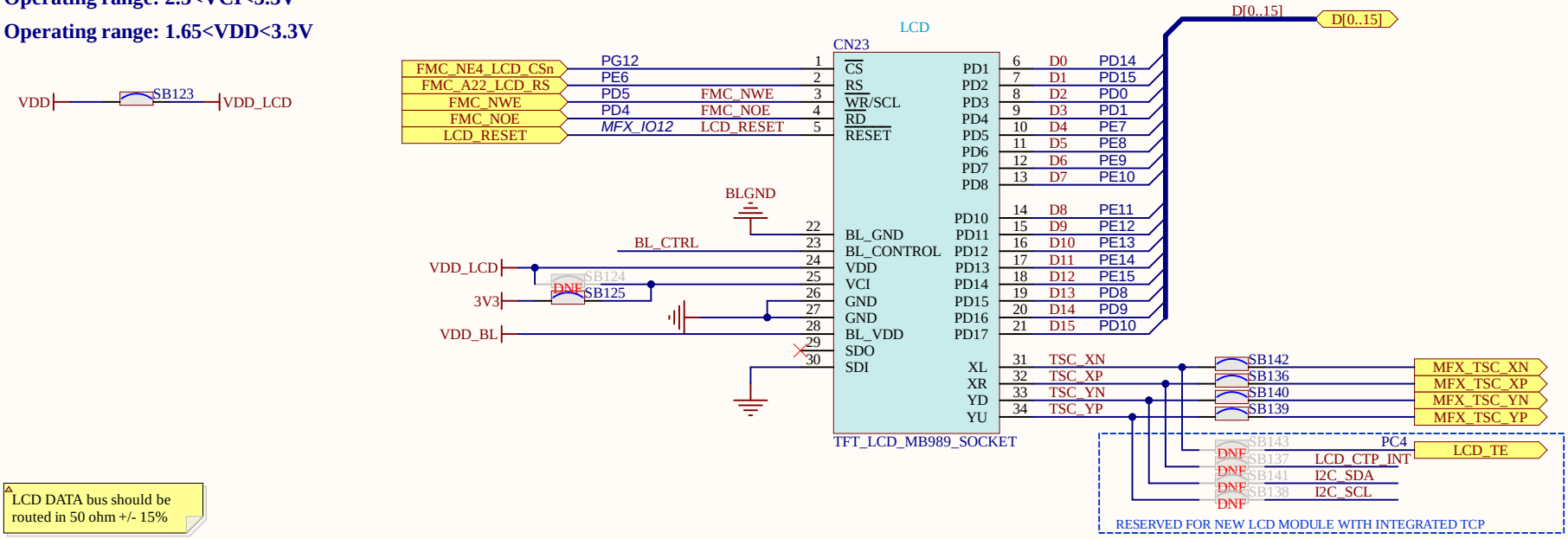
Title: **MFX IO EXPANDER**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 6 of 24

life.augmented

TFT LCD SOCKET CONNECTOR

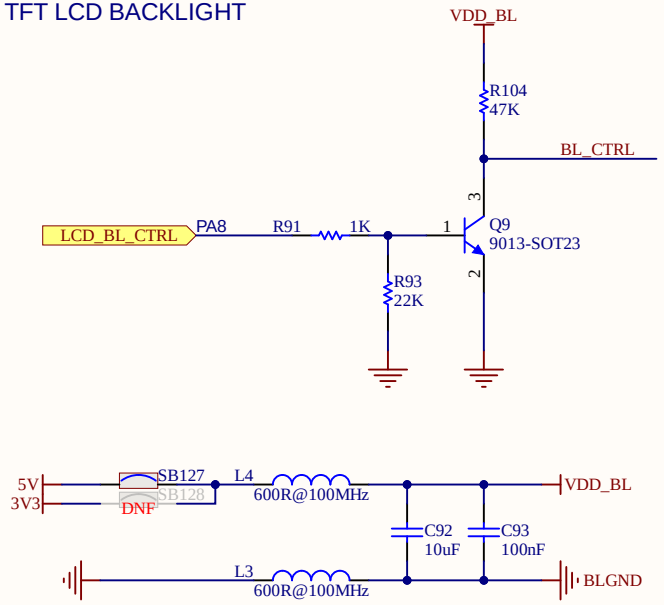
Operating range: 2.5<VCI<3.3V

Operating range: 1.65<VDD<3.3V



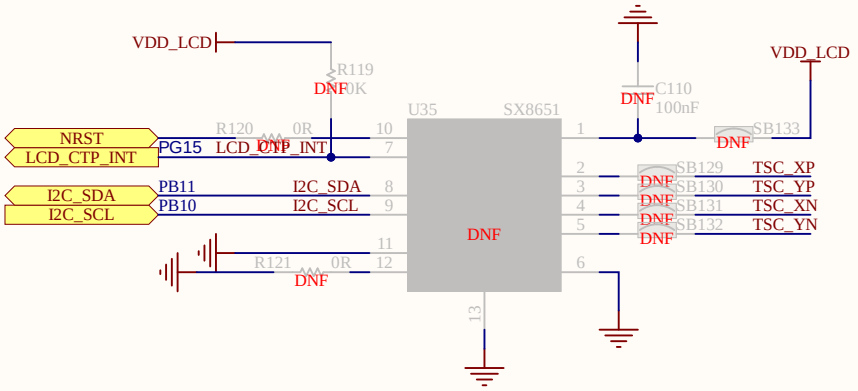
LCD DATA bus should be routed in 50 ohm +/- 15%

TFT LCD BACKLIGHT



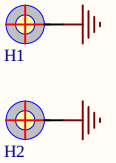
TOUCH PANEL SEMTECH

Operating range: 1.65<VDD<3.7V



Default I2C Address:0x90: 0b:1001.000x

LCD MECHANICAL PARTS



LIMITATIONS : See User Manual for more detail about muxing.

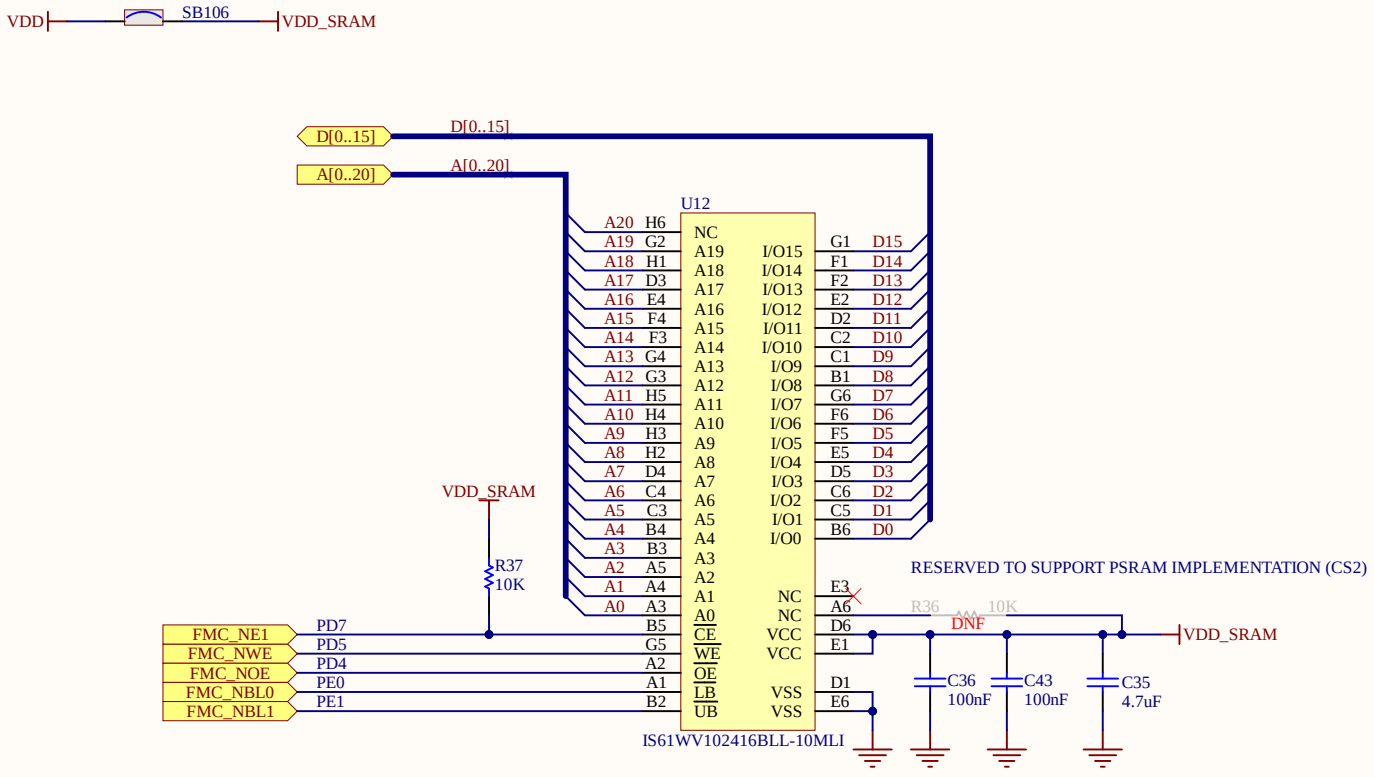
LCD_TE exclusive with SMARTCARD_UART_RX

Title: **LCD TFT 320x240**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 7 of 24

life.augmented

SRAM

Operating range: 2.4<VDD<3.6V

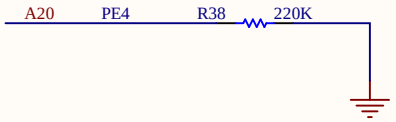


Other memory compatible:
SRAM 512kx16: IS61WV51216BLL-10MLI
PSRAM 512Kx16: IS66WV51216EBLL-55BLI

Place CAPS close to SRAM PWR pin

Default config = A0-A19

A20 = 0 by default (depend of the memory size)



SRAM bus should be routed in 50 ohm +/- 15%

LIMITATIONS : See User Manual for more detail about muxing.

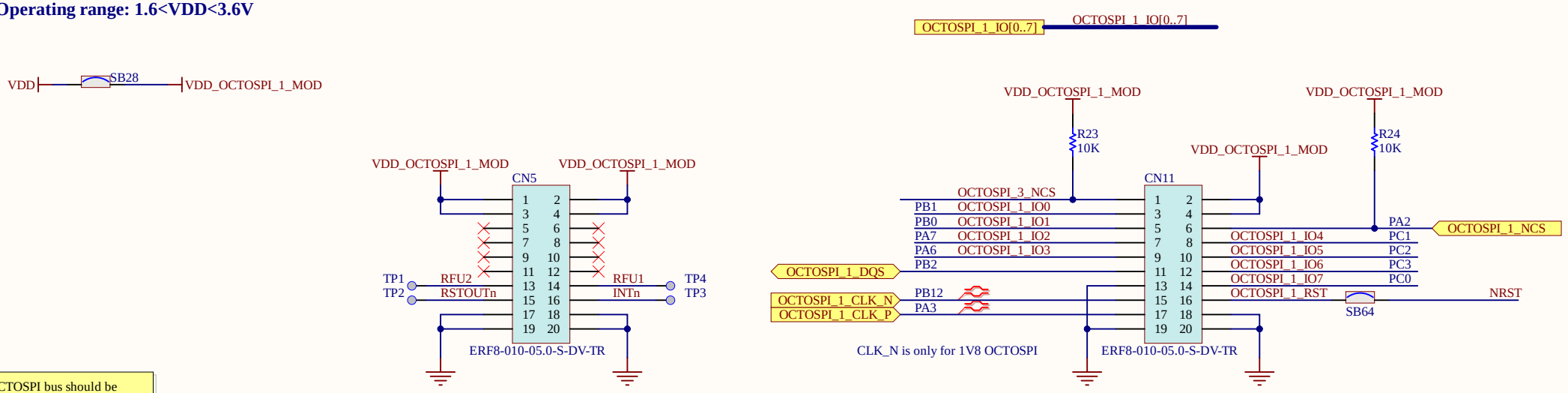
Memory size limited to max 32Mbits by design

Title: **SRAM**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 8 of 24

life.augmented

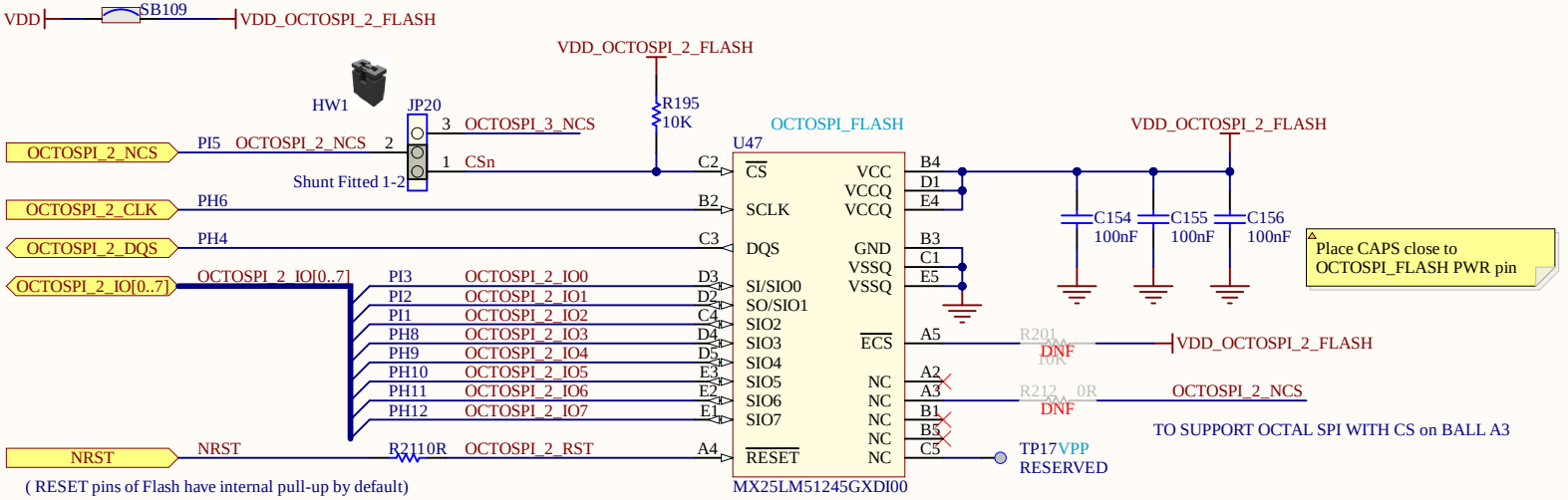
OCTOSPI_1 (MB1242 Module connector)

Operating range: 1.6<VDD<3.6V



OCTOSPI_2 FLASH

Operating range: 2.7<VDD<3.6V



LIMITATIONS : See User Manual for more detail about muxing.

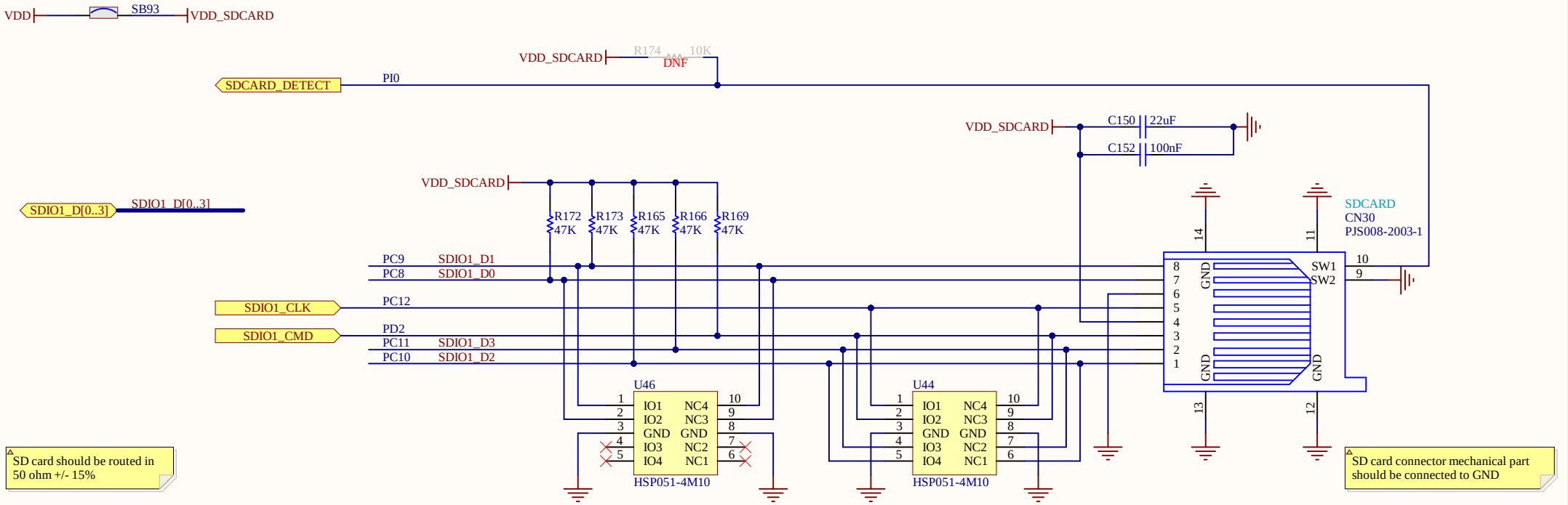
OCTOSPI FLASH IO exclusive with MC.
OCTOSPI_FLASH_CLK exclusive with OPAMP
OCTOSPI SRAM IO exclusive with CAMERA.

Title: **OCTOSPI**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 9 of 24

life.augmented

SD CARD

Operating range: 2.7<VDD<3.6V



LIMITATIONS : See User Manual for more detail about muxing.

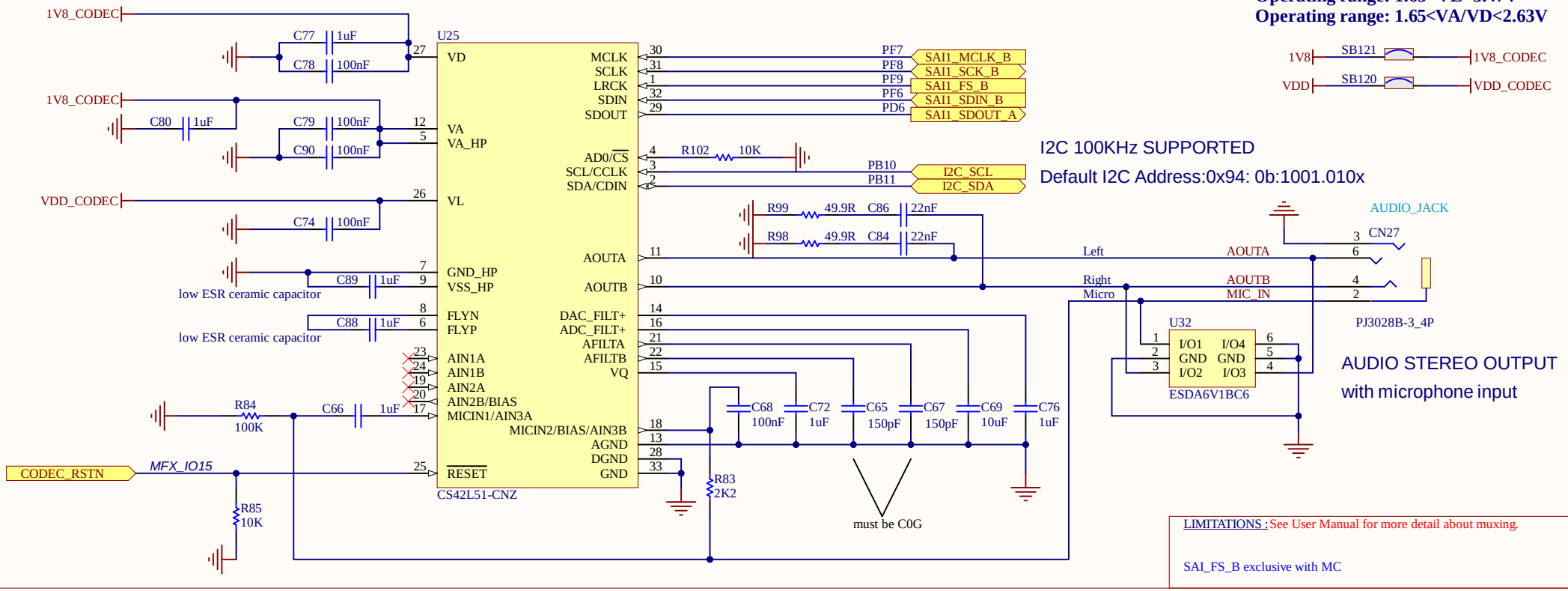
SDIO_CK/D1/D2 exclusive with TRACE_D0/D1/D3
SDIO_D0 exclusive with MC
SDIO_CLK exclusive with STMOD+

Title: **SDCARD**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 10 of 24

life.augmented

AUDIO CODEC

Operating range: 1.65<VL<3.47V
Operating range: 1.65<VA/VD<2.63V

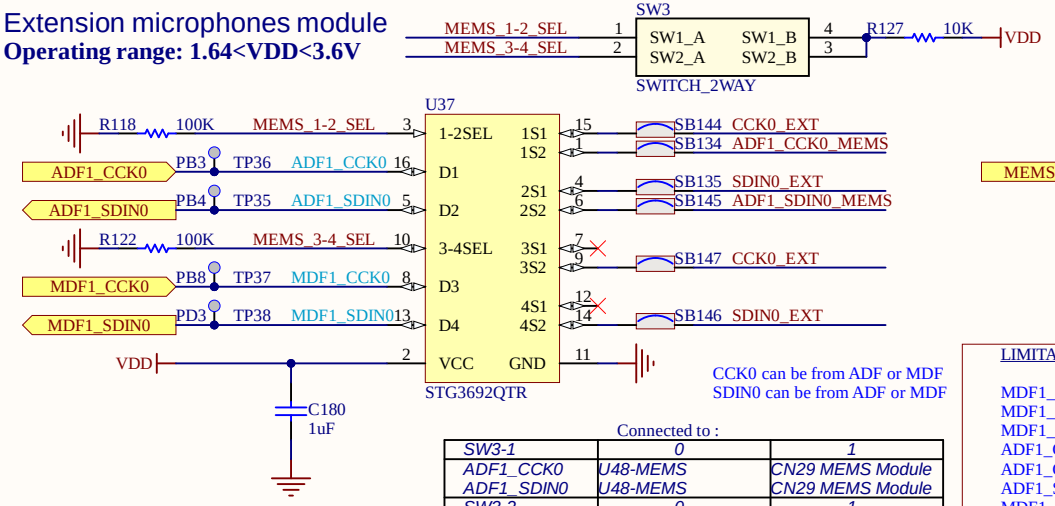


LIMITATIONS : See User Manual for more detail about muxing.

SAI_FS_B exclusive with MC

Extension microphones module

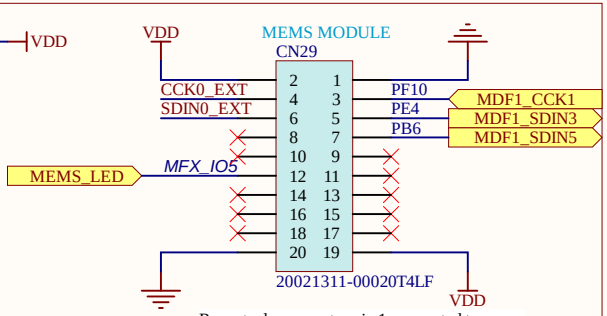
Operating range: 1.64<VDD<3.6V



Connected to :

SW3-1	0	1
ADF1_CCK0	U48-MEMS	CN29 MEMS Module
ADF1_SDIN0	U48-MEMS	CN29 MEMS Module
SW3-2	0	1
MDF1_CCK0	CN29 MEMS Module	NC
MDF1_SDIN0	CN29 MEMS Module	NC

SEL = 0 : Dx connected to xS2
SEL = 1 : Dx connected to xS1

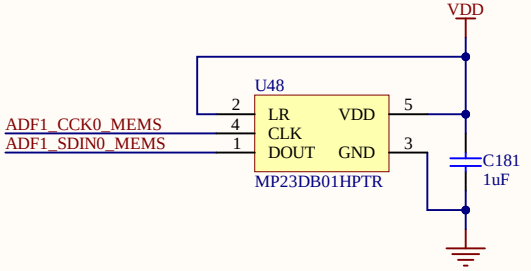


LIMITATIONS : See User Manual for more detail about muxing.

MDF1_CCK1 shared between EXT-MEMS, STPMS2, STMOD+
MDF1_CCK1 exclusive with MC
MDF1_SDIN3 shared between EXT-MEMS, STPMS2, STMOD+
ADF1_CCK0 shared between EXT-MEMS and INT-MEMS
ADF1_CCK0 exclusive with JTAG-JTDO
ADF1_SDIN0 exclusive with JTAG-NJTRST, COMP2 and STMOD+
MDF1_SDIN5 shared EXT-MEMS, STPMS2, STMOD+
MDF1_SDIN5 exclusive with MC
MDF1_CCK0 exclusive with UCPD, FDCAN

MEMS

Operating range: 1.64<VDD<3.6V

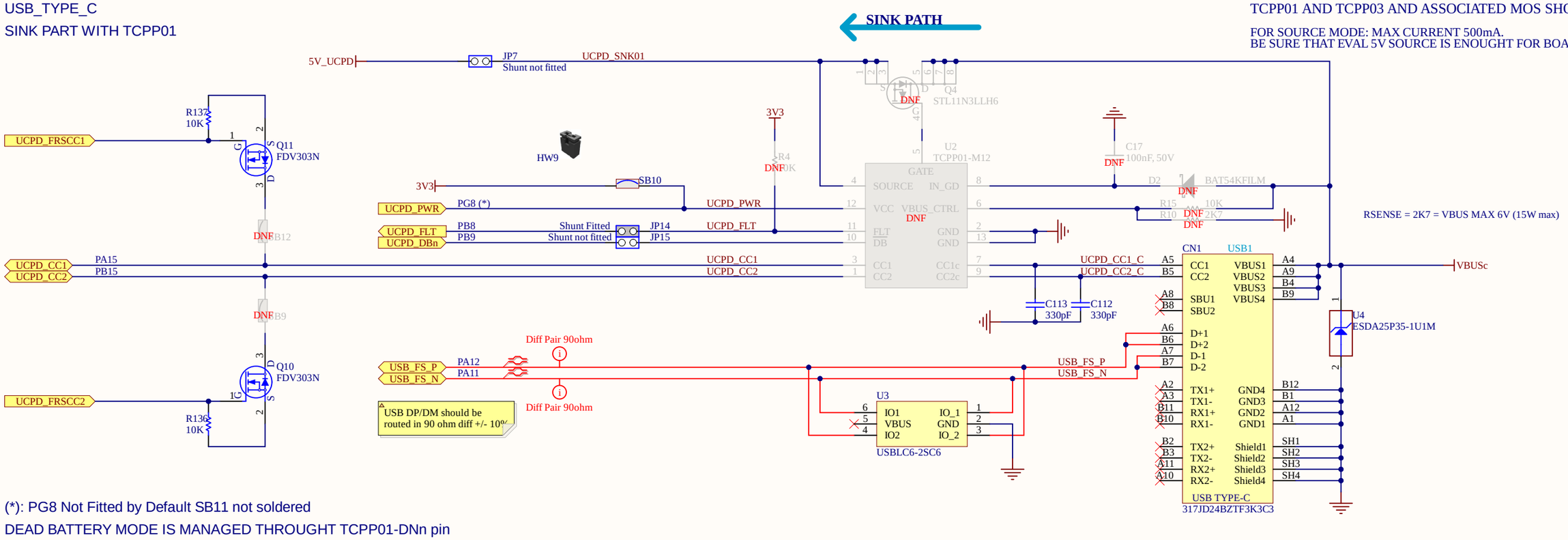


Title: **AUDIO**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 11 of 24

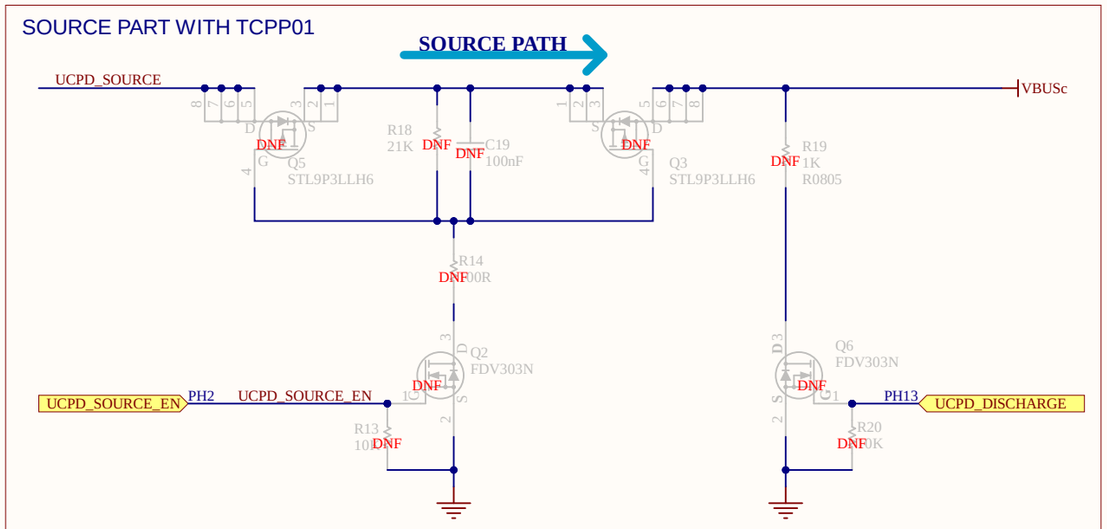
life.augmented

USB_TYPE_C

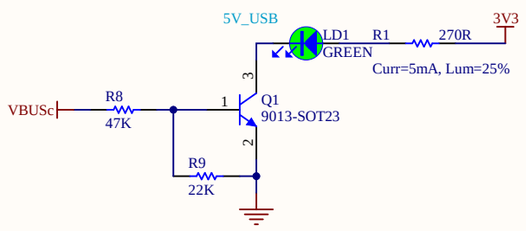
SINK PART WITH TCPP01



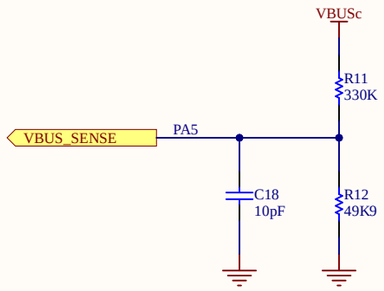
SOURCE PART WITH TCPP01



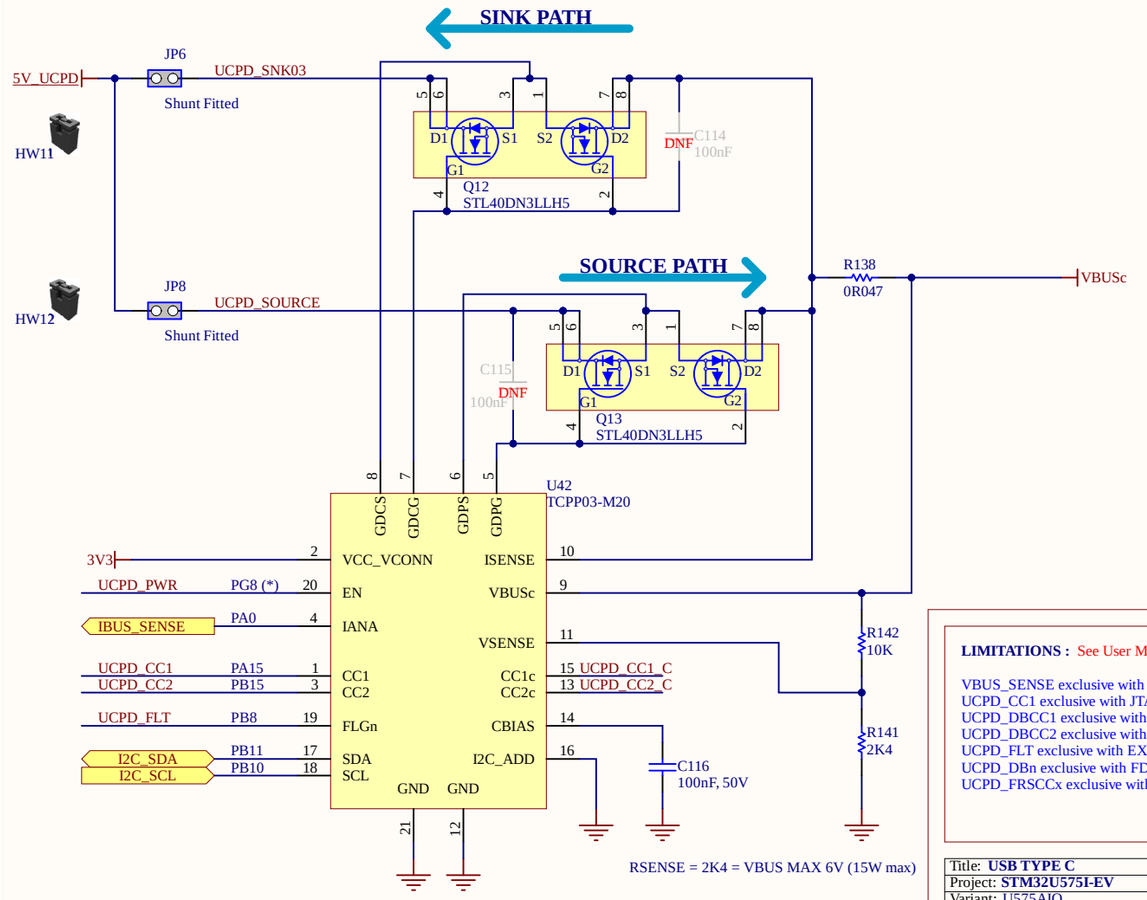
5V_USB_LED



VBUS SENSE



UCPB MANAGEMENT WITH TCPP03



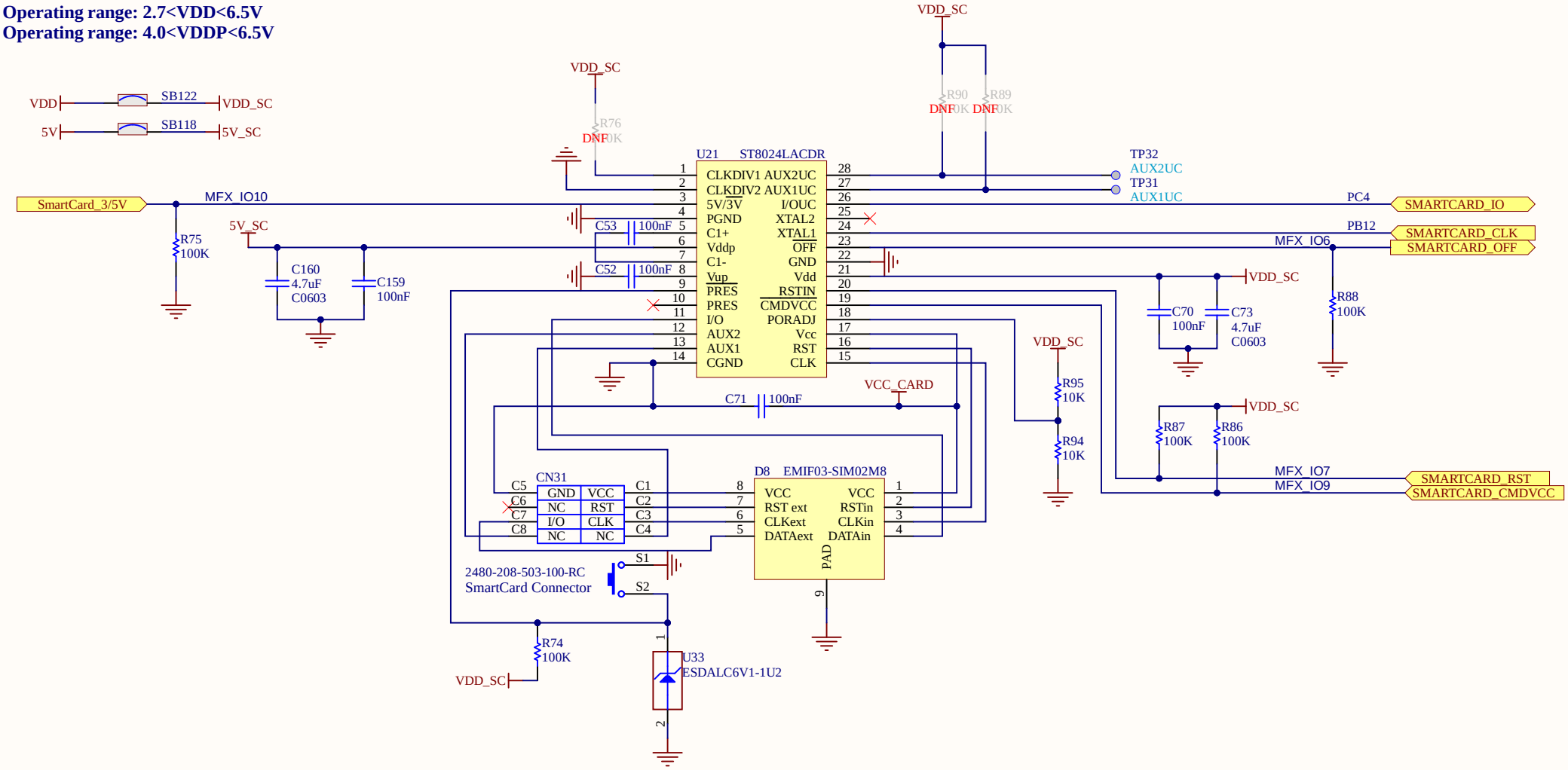
LIMITATIONS : See User Manual for more detail about muxing.

VBUS_SENSE exclusive with STMOD+
UCPD_CC1 exclusive with JTAG_TDI
UCPD_DBCC1 exclusive with COMP and STMOD+
UCPD_DBCC2 exclusive with TOUCH-KEY
UCPD_FLT exclusive with EXT-MEMS, FDCAN
UCPD_DBn exclusive with FDCAN
UCPD_FRSCCx exclusive with RS232 and STMOD+ (UART)

Title: **USB TYPE C**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A3	Date: 02 Sept 2021	Sheet: 12 of 24

SmartCard

Operating range: 2.7<VDD<6.5V
Operating range: 4.0<VDDP<6.5V



LIMITATIONS : See User Manual for more detail about muxing.

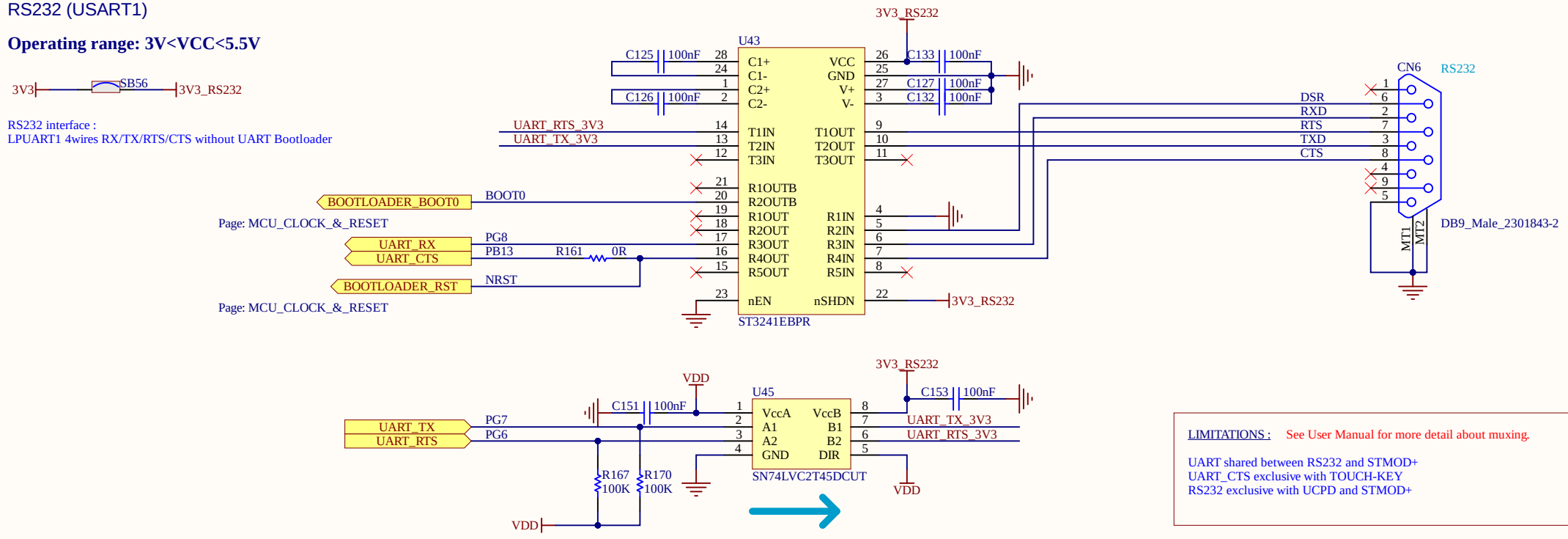
SMARTCARD_UART_RX exclusive with LCD_TE
SMARTCARD_CLK exclusive with MC

Title: **SMARTCARD**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 13 of 24

life.augmented

RS232 (USART1)

Operating range: 3V<VCC<5.5V

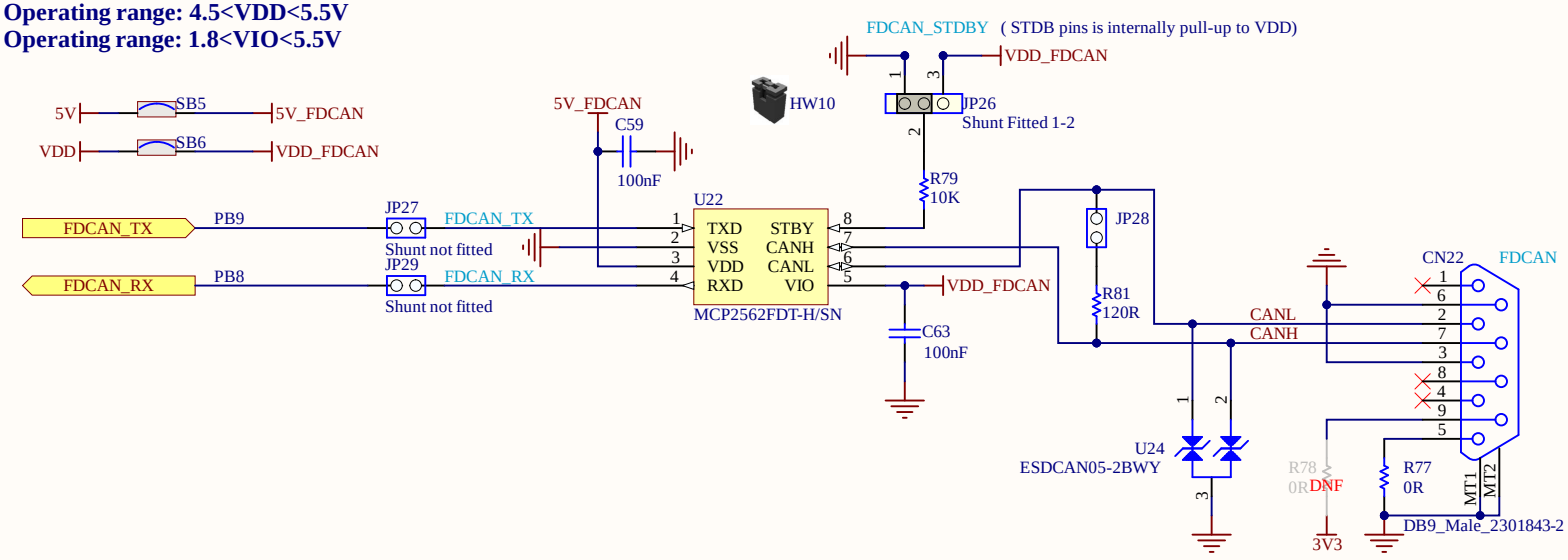


LIMITATIONS : See User Manual for more detail about muxing.

UART shared between RS232 and STMOD+
UART_CTS exclusive with TOUCH-KEY
RS232 exclusive with UCPD and STMOD+

CAN FD

Operating range: 4.5<VDD<5.5V
Operating range: 1.8<VIO<5.5V



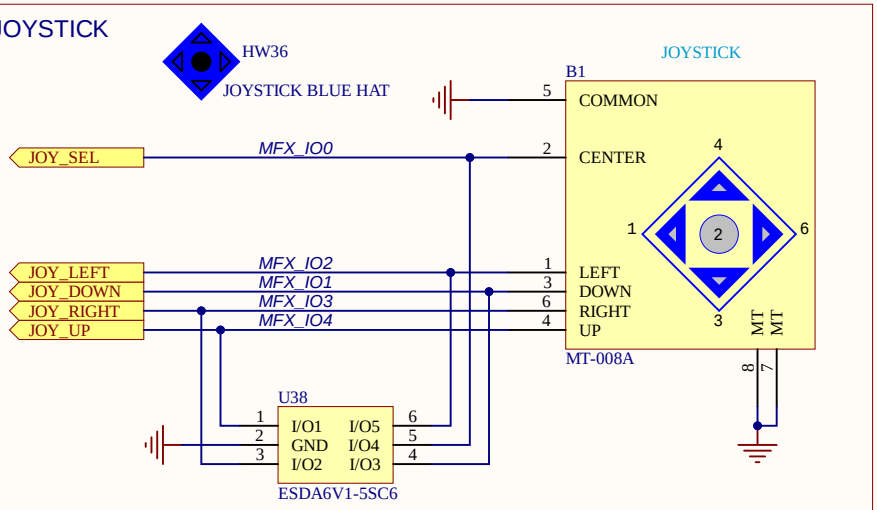
LIMITATIONS : See User Manual for more detail about muxing.

FDCAN_TX exclusive with UCPD_DBn
FDCAN_RX exclusive with EXT-MEMS, UCPD

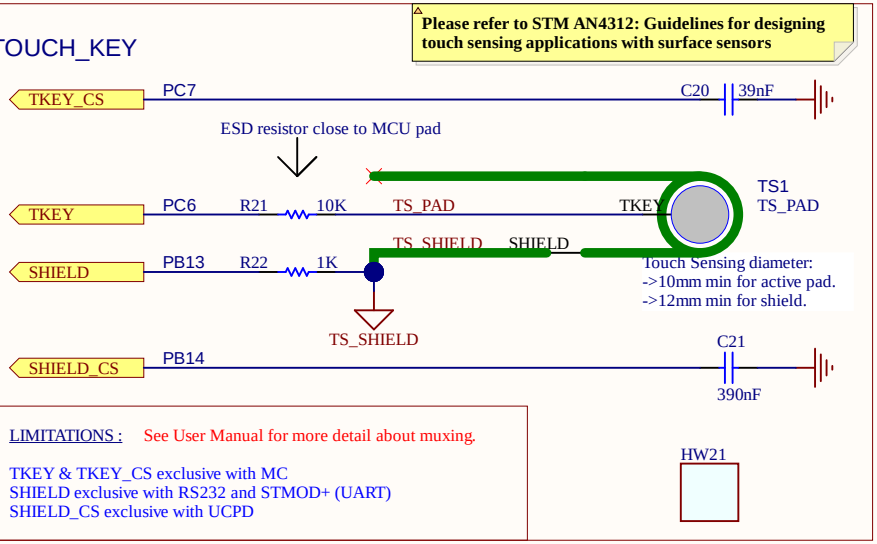
Title: **FDCAN & RS232**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 14 of 24

life.augmented

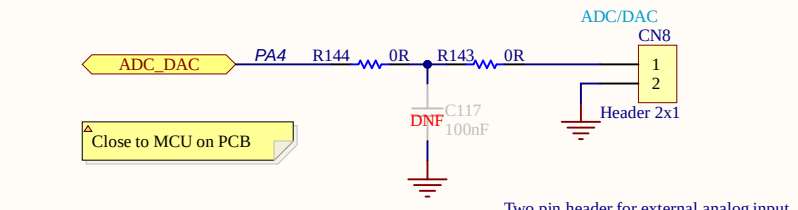
JOYSTICK



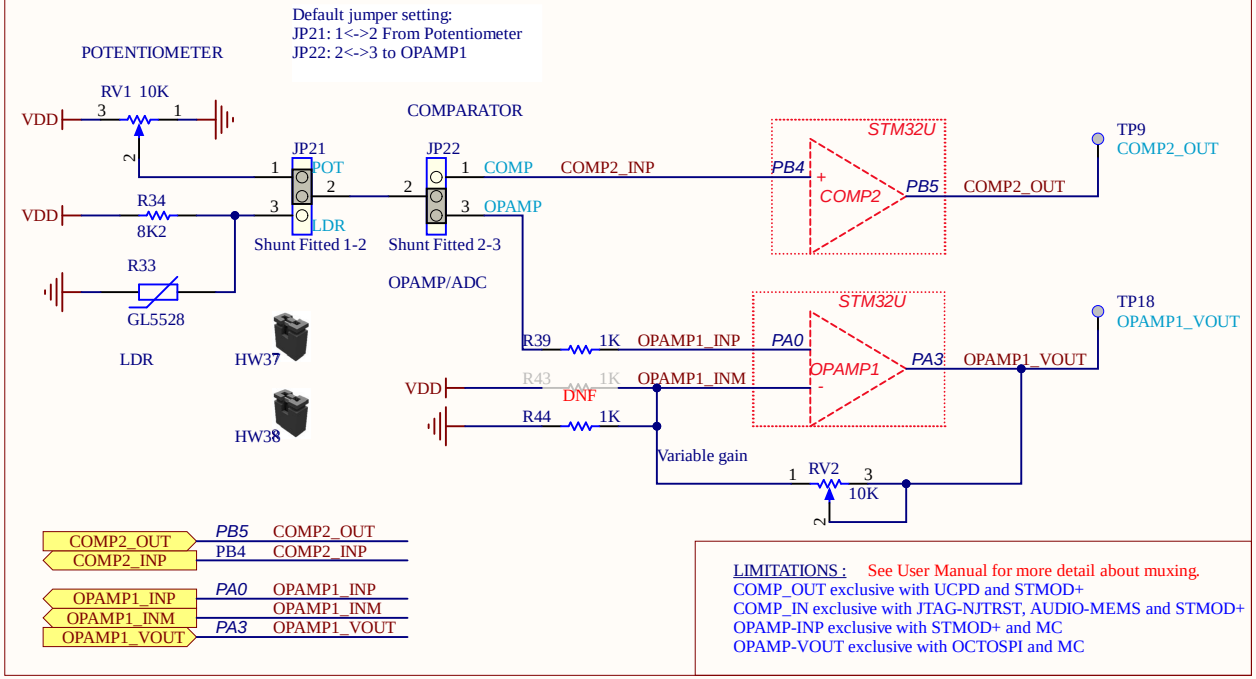
TOUCH_KEY



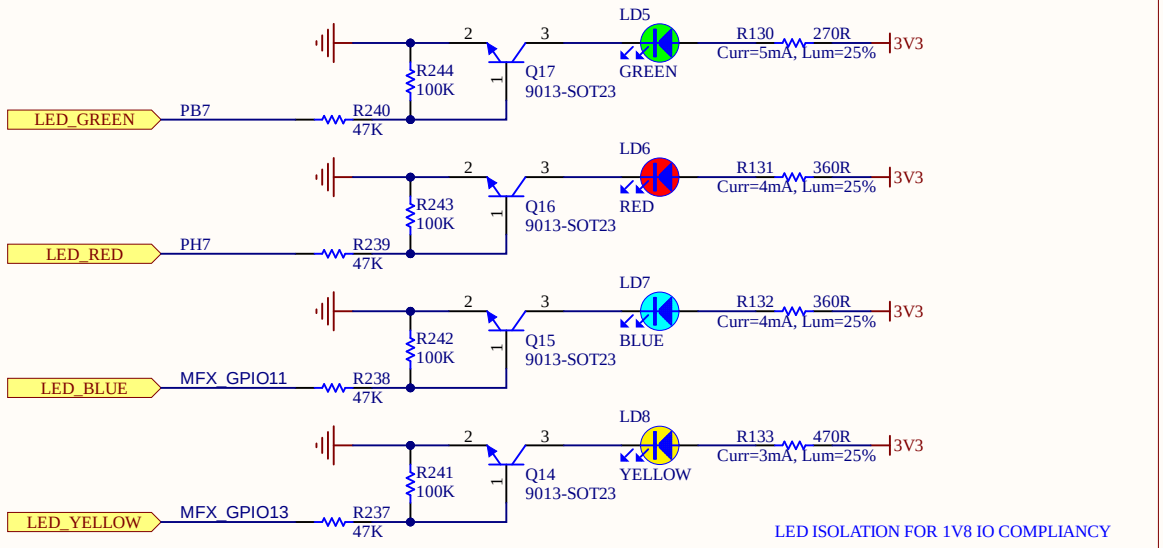
ADC / DAC CONNECTOR



LDR / POTENTIOMETER



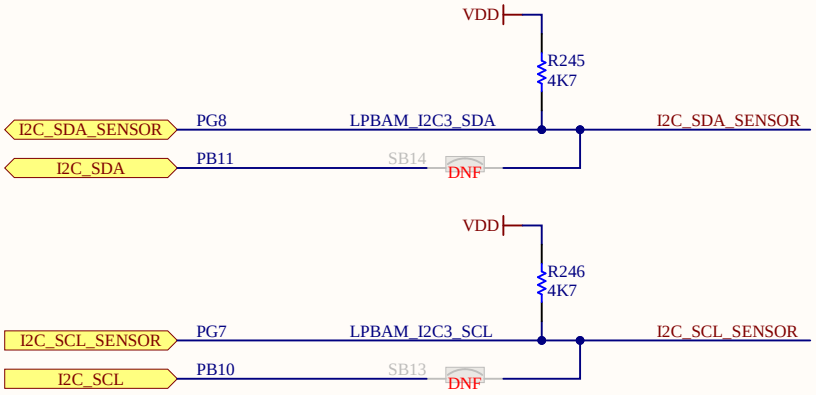
LEDs



Title: **PERIPHERAL**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 15 of 24

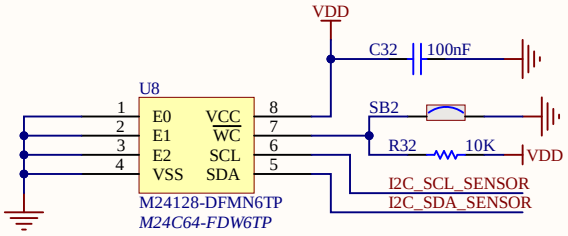
life.augmented

I2C SELECTION



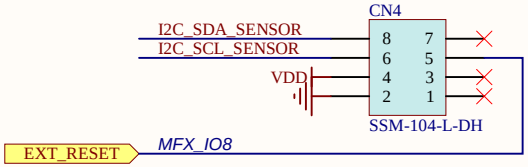
I2C EEPROM

Operating range: 1.7V<VCC<5.5V



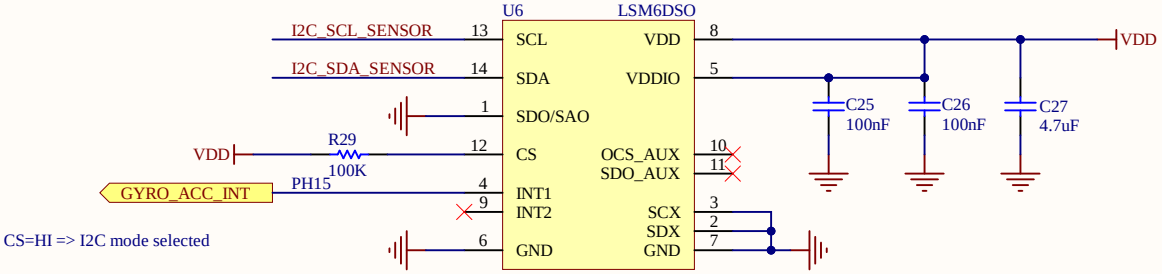
Default I2C Address:0xA0: 0b:1010.000x

EXT_I2C



3D ACCELEROMETER and 3G GYROSCOPE SENSOR

Operating range: 1.71V<VCC<3.6V



CS=HI => I2C mode selected

Default I2C Address: Read 0xD5 / Write 0xD4

LIMITATIONS : See User Manual for more detail about muxing.

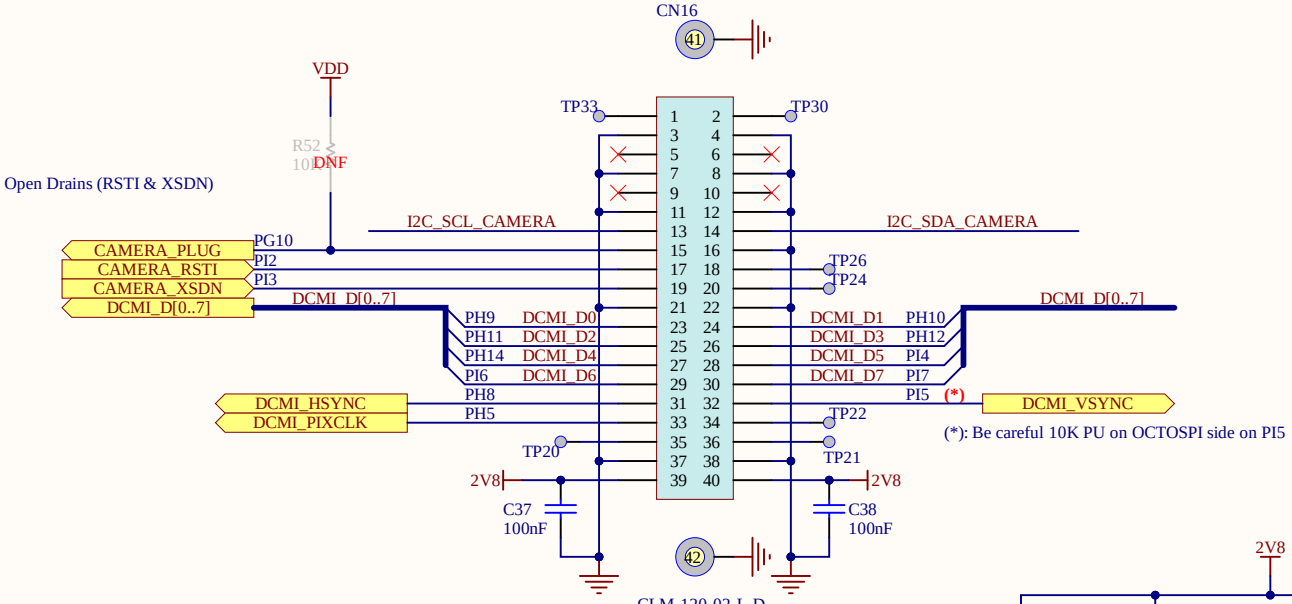
NO LIMITATION

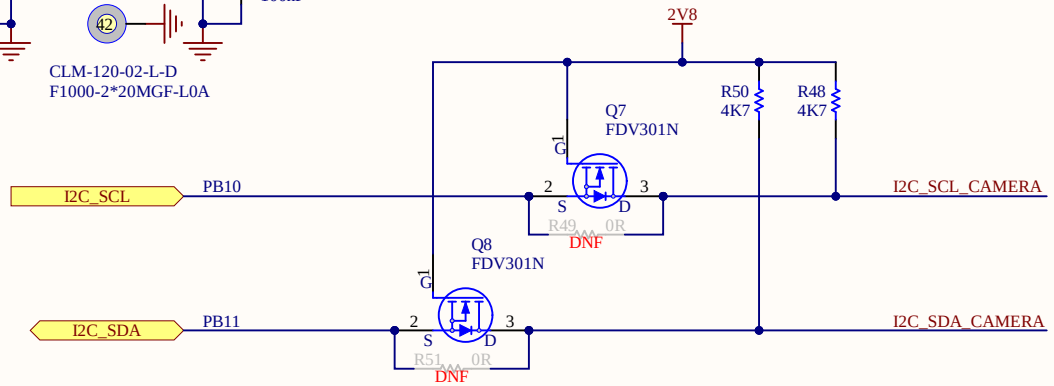
Title: **SENSOR & I2C PERIPHERALS**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 16 of 24

life.augmented

CAMERA MODULE CONNECTOR

Operating range: 2.8<VDD<3.6V





I2C level Shifter for MCU @3V3 to Camera @2V8

Default I2C Address: OV5640_DeviceWriteAddress 0x78

OCTOSPI bus should be routed in 50 ohm +/- 15%

LIMITATIONS : See User Manual for more detail about muxing.

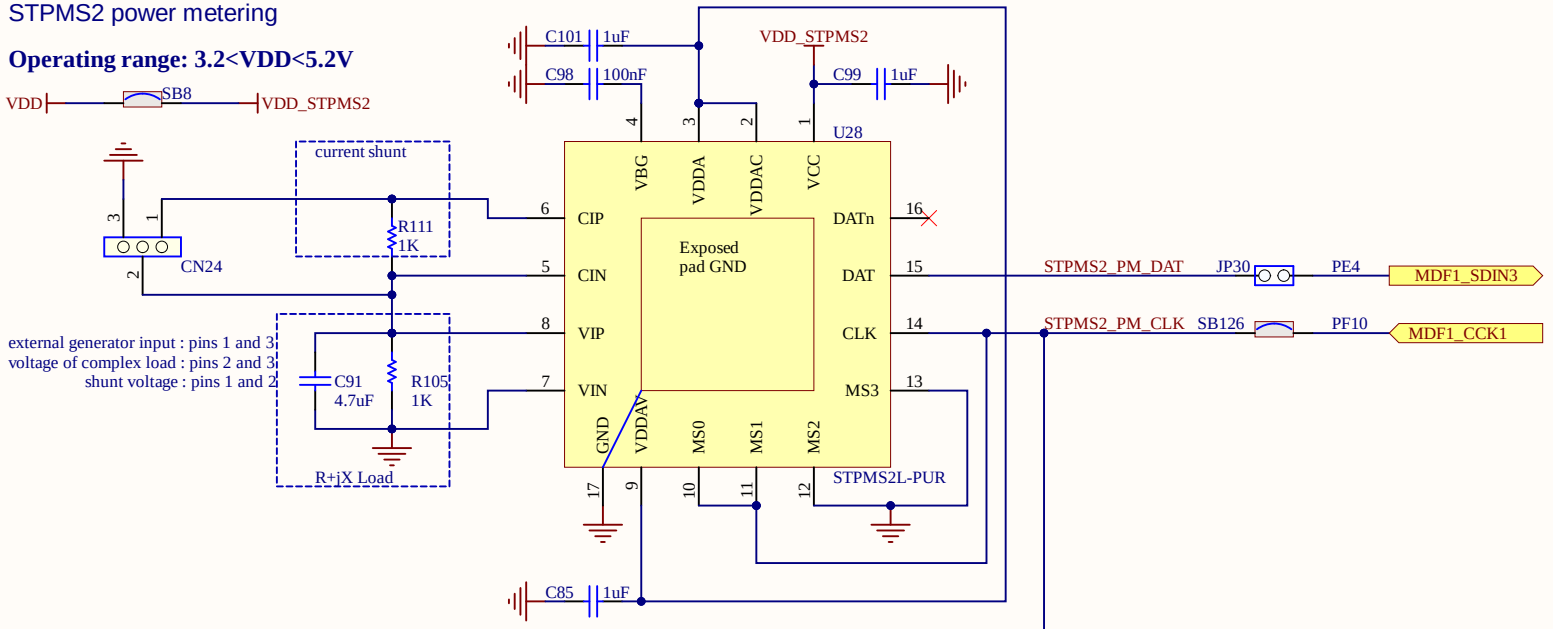
CAMERA interface exclusive with OCTOSPI SRAM

Title: **CAMERA**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 17 of 24

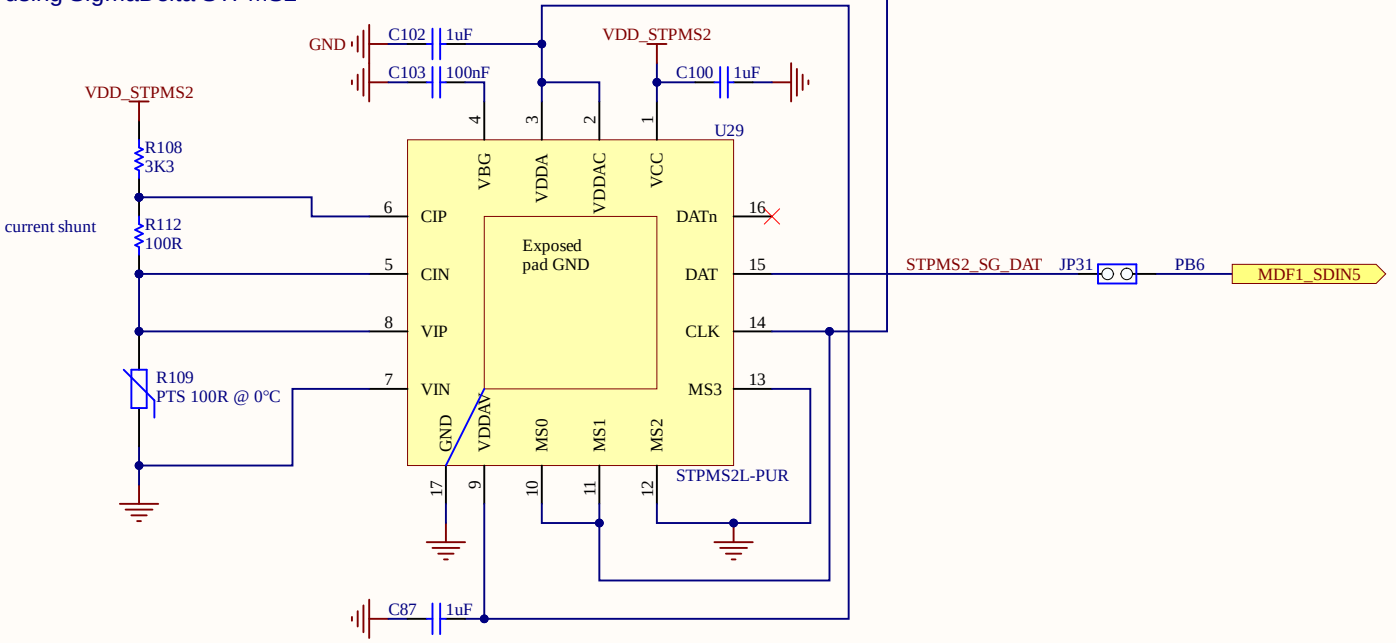
life.augmented

STPMS2 power metering

Operating range: 3.2<VDD<5.2V



Measurement using SigmaDelta STPMS2



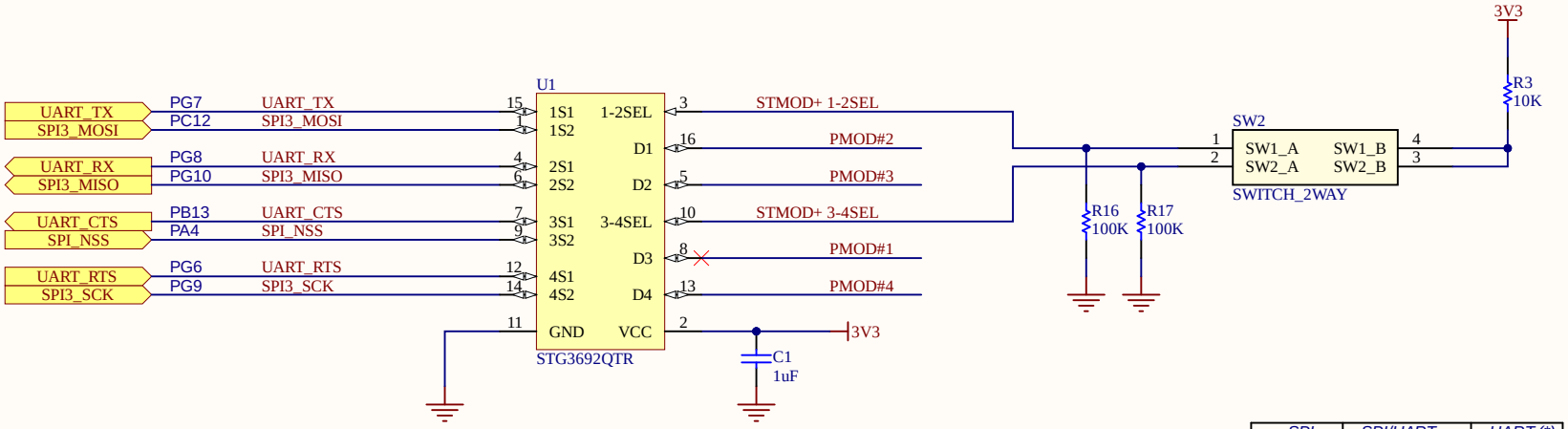
LIMITATIONS : See User Manual for more detail about muxing.

MDF1_SDIN3 shared between EXT-MEMS, STPMS2, STMOD+
MDF1_CCK1 shared between EXT-MEMS, STPMS2, STMOD+
MDF1_CCK1 exclusive with MC
MDF1_SDIN5 shared between EXT-MEMS, STPMS2, STMOD+

Title: **STPMS2**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 18 of 24

life.augmented

PMOD / STMOD+ SWITCH

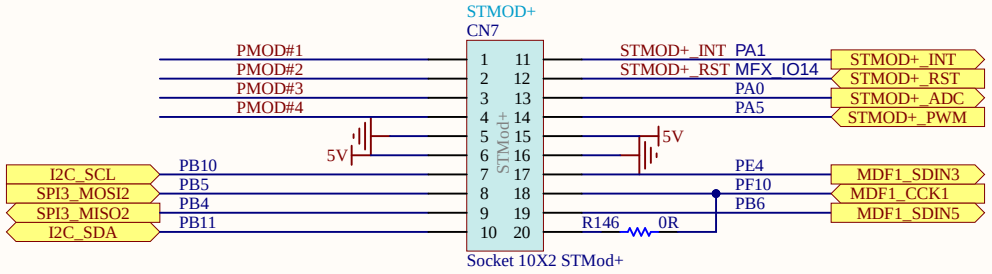


SEL = 1 : Dx connected to xS1
SEL = 0 : Dx connected to xS2

	SPI	SPI/UART	UART (*)
STMOD+ 1-2SEL	0	1	1 (*)
STMOD+ 3-4SEL	0	0	1 (*)
PMOD#1	NSS	NSS	CTS
PMOD#2	MOSIp	TX	TX
PMOD#3	MISOp	RX	RX
PMOD#4	SCLK	SCLK	RTS

(*) default configuration to support MikroBus using STMOD+ MB1280 fan-out board

STMOD+



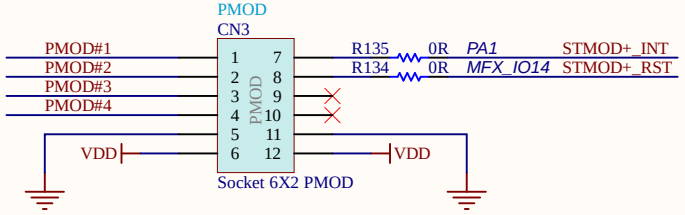
LIMITATIONS : See User Manual for more detail about muxing.

SPI exclusive with VBUS_SENSE, ADC/DAC, TRACE, SDMMC and MC
UART exclusive with TOUCH-KEY, RS232 interface and UCPD

SPI3-MOSI2 exclusive with COMP2 and UCPD
SPI3-MISO2 exclusive with int-MEMS, JTAG and COMP

INT shared between STMOD+ and PMOD
INT exclusive with TAMPER and MC
RST shared between STMOD+ and PMOD
STMOD+_ADC exclusive with OPAMP and MC
STMOD+ PWM exclusive with VBUS_SENSE
MDF1_CCK1 shared between EXT-MEMS, STPMS2, STMOD+
MDF1_CCK1 exclusive with MC
MDF1_SDIN3 shared between EXT-MEMS, STPMS2, STMOD+
MDF1_SDIN5 shared between EXT-MEMS, STPMS2, STMOD+
MDF1_SDIN5 exclusive with MC

PMOD



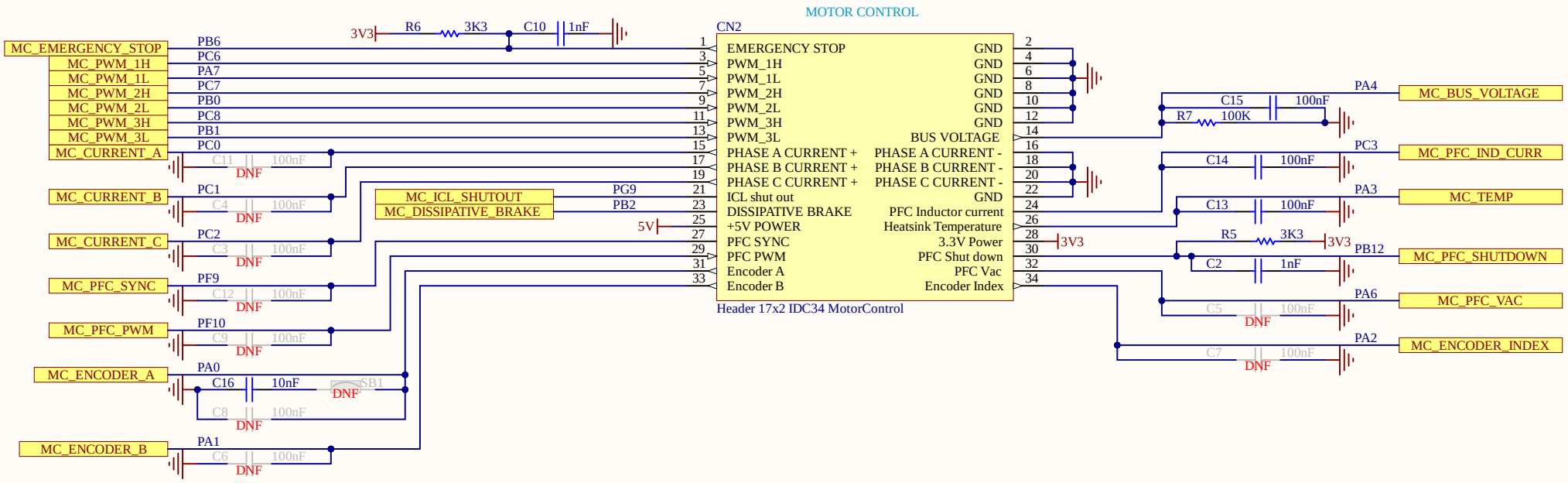
LIMITATIONS : See User Manual for more detail about muxing.

RST shared between STMOD+ and PMOD
INT shared between STMOD+ and PMOD
PMOD/STMOD+ INT exclusive with TAMPER and MC
SPI exclusive with VBUS_SENSE, ADC/DAC, TRACE, SDMMC and MC
PMOD/STMOD+ UART exclusive with TOUCH-KEY, RS232 interface and UCPD

Title: **PMOD & STMOD+**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 19 of 24

life.augmented

MOTOR CONTROL



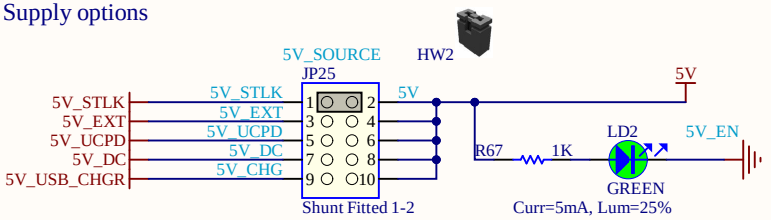
LIMITATIONS : See User Manual for more detail about muxing.

MotorControl exclusive with many functions.
See UM for MC IO configuration

Title: **MOTOR CONTROL**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 20 of 24

life.augmented

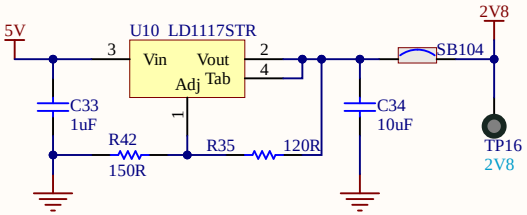
5V Power Supply options



MAX CURRENT 500mA WHEN USING POWER INPUT 5V_STLK

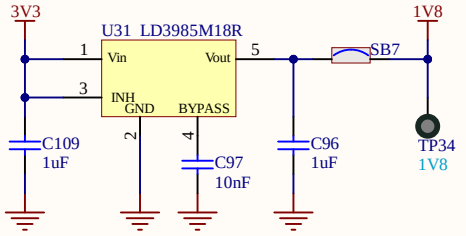
2V8_LDO: CAMERA

2V8 / 800mA



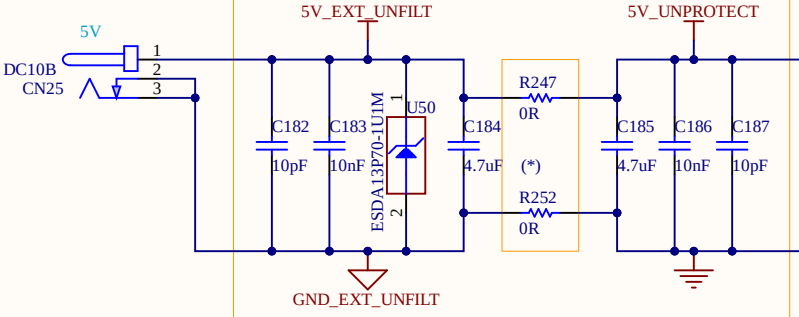
1V8_LDO: AUDIO_CODEC

1V8 / 150mA



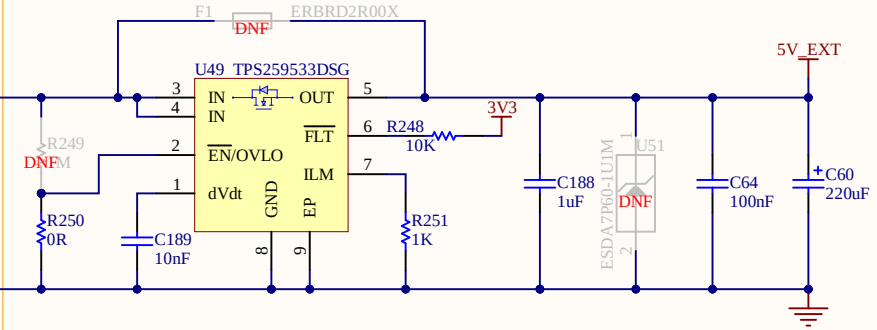
5V External supply

5V_EXT: ESD / EMI Filtering

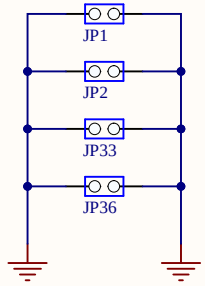


(*) footprint compatible with DLW5BTM142TQ2L

5V_EXT: OVP & OCP Protection (max 2A, 5.7V)

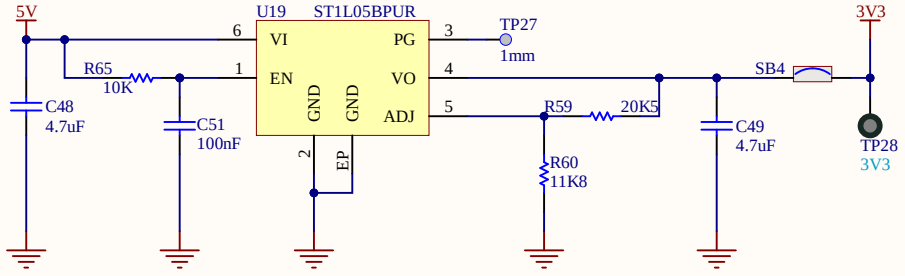


GND TEST POINT



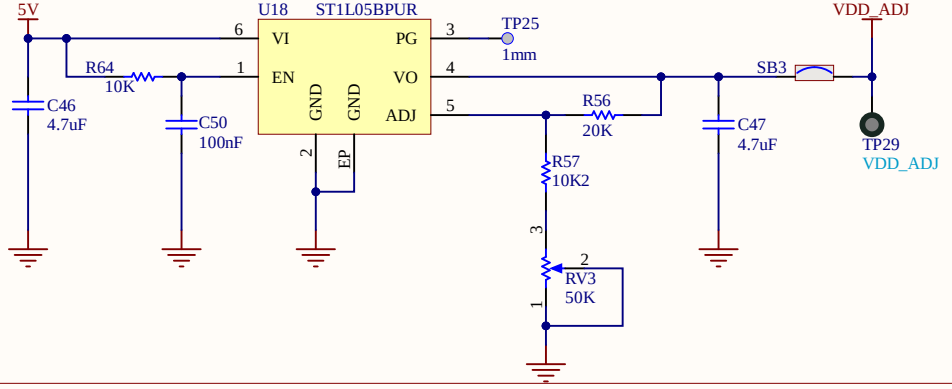
3V3 Supply

3V3 / 1.3A
Vout=1.22(1+R1/R2)*



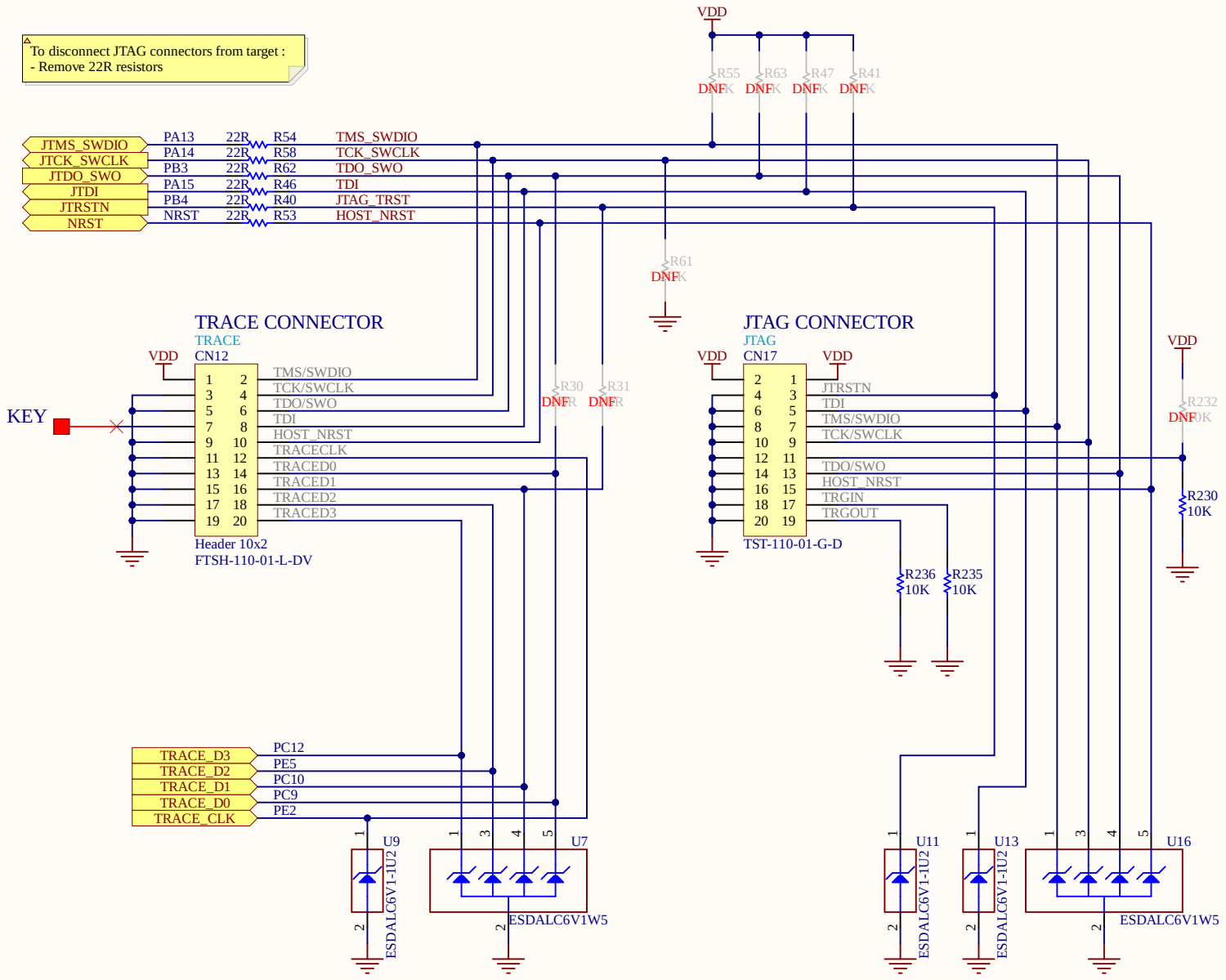
VDD_ADJ Supply

VDD_ADJ [1.7V to 3.61V] / 1.3A
Vout=1.22(1+R1/R2)*



Title: **BOARD PWR**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 21 of 24

JTAG & TRACE



LIMITATIONS : See User Manual for more detail about muxing.

TRACE_D0/D1/D3 exclusive with SDIO_CK/D1/D2
TRACE_D3 exclusive with STMOD+-SPI
JTAG_TDI exclusive with UCPD_CC1
JTAG-JTDO exclusive with ADF1_CCK0
JTAG-NJTRST exclusive with ADF1_SDIN0, COMP2 and STMOD+ SPI

Title: **JTAG & TRACE**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 22 of 24

life.augmented

STLINK_MCU

JTCK_SWCLK — PA14 — JTCK_SWCLK
JTMS_SWDIO — PA13 — JTMS_SWDIO
JTDO_SWO — PB3 — JTDO_SWO
JTDI — PA15 — JTDI
T_VCP_TX — PA9 — T_VCP_TX
T_VCP_RX — PA10 — T_VCP_RX

NRST — NRST
STLK_MCO — STLK_MCO

JTRSTN — PB4 — JTRSTN

LIMITATION :

Some JTAG signals are multiplexed with UCPD: JTDI
In case JTDI is plugged, USB TYPE C connector should be disconnected.
User need to check that multiplexed signals are not used on both features simultaneously.

The routing of the tracks must be done on continuity from ST-LINK to 22ohm resistors and from resistors to Target MCU to avoid stub noises and EMC approach. (22ohm resistors must be very close to STDC14)

CN28 RES — DNF — Header 4x1

Connector must be on the border of the PCB

3V3_STLK

STLK_SWDIO — A15
STLK_SWCLK — A14

HSE CLK STLINK

3V3_STLK — X3 — 4 VDD OUT 3 — R97 100R — G1
R92 10K — 1 EN GND 2 — NZ2520SH 25MHz — H1
C75 100nF

STLK_NRST — JP32 — C83 100nF — J1

HW6 DNF — D6

3V3_STLK — C6

C106 2.2uF — R115 3K — J14 J15

USB_DEV_HS_N — R14
USB_DEV_HS_P — R15

U26A

STLK_SWDIO/PA13
STLK_SWCLK/PA14
USB_DEV_HS_N/PB14
USB_DEV_HS_P/PB15
OSC_IN/PH0
OSC_OUT/PH1
NRST
BOOT0
PDR_ON
V12_OTGPHY
REXT_OTGPHY

T_SW_DIR/PA7 — R3
T_SWDIO_IN/PH7 — N12 — T_SWDIO_IN — level shifter 1V8/3V3
T_SWDIO_OUT/PF9 — L2 — T_SWDIOa — level shifter 1V8/3V3
T_SWCLK/PH6 — M11 — T_SWCLKa
T_SWO/PD2 — D12 — T_SWOa — level shifter 1V8/3V3
STLK_VCP_RX/PG9 — C10 — STLK_VCP_RX — level shifter 1V8/3V3
STLK_VCP_TX/PG14 — A7 — STLK_VCP_TX
GNDDetect/PG5 — K13 — GNDDetect
T_VCC AIN/PA0 — N3
T_PWR_EXT/PB1 — R4 — T_PWR_EXT
T_PWR_EN/PB0 — R5 — T_PWR_EN
STLK_LED/PA10 — D15 — LED_STLK
STLK_MCO/PA8 — F15
T_NRST/PA6 — P3 — T_NRSTa
HW_TYP_0/PI4 — D4
HW_TYP_1/PI5 — C4

4K7 R100 — 3V3
4K7 R101
2K7 R106 — 5V
4K7 R107

C108 20pF DNF — R116 100R DNF — STLK_MCO

ST-LINKV3E_SWD

ST-LINK POWER

3V3_STLK — U26B

VDD: N10, G13, C5, K4, G3, N8, N9, J13, H13, C8, C7, F3, J12, C9, R1
VDDA: P1
VREF+: C1
VBAT: L4
BYPASS-REG: M10
VCAP1: F13
VCAP2
VSS: F6, F7, F8, F9, F10, G6, G7, G8, G9, H6, H7, H8, H9, H10, J6, J7, J8, J9, J10, K6, K7, K8, K9, K10, F12, D5, G2, M8, M1, N1, G10, D9, H12, F2, D7, D8, G12, M9
VSSA
VREF-

R96 10K
C171 2.2uF — C177 2.2uF

ST-LINKV3E_SWD

Ceramic capacitor (Low ESR)

3V3_STLK
C174 100nF — C178 100nF — C162 100nF — C165 100nF — C167 100nF — C169 100nF — C172 100nF — C175 100nF

3V3_STLK
C176 100nF — C170 100nF — C168 100nF — C164 100nF — C179 100nF — C173 100nF — C166 100nF — C161 100nF

3V3_STLK
C81 4.7uF — C82 1uF — C163 100nF

TAG CONNECTOR

VDD
CN13 TAG — DNF — TC2050-IDC-NL

JTMS_SWDIO PA13 — NRST
JTCK_SWCLK PA14 — PB4 JTRSTN
PA15 JTDI
PB3 JTDO_SWO

TAG: Only footprint with cable: TC2050-IDC-NL

ESD PROTECTIONS

U17 — ESDALC6V1W5
JTMS_SWDIO 5
JTCK_SWCLK 4
JTDO_SWO 3
JTDI 1
2

U14 — ESDALC6V1W5
STLK_SWDIO 5
STLK_SWCLK 4
T_VCP_TX 3
T_VCP_RX 1
2

Pin attributions on ESD can be swapped for layout optimisation

LED STLK

LD4 COM
Red
Green
LD_BICOLOR_CMS

1 — R128 330R — 2 — LED_STLK
4 — R129 330R — 3 — 3V3_STLK

STDC14 RECEIVER

CN18 — MIPI10 on STDC14 footprint — MIPI10

VDD
KEY
GNDDetect
T_VCP_RX PA10

R117 100R — STLK_VCP_TX
R68 47R — T_VCP_RX

1	2	PA13	JTMS_SWDIO		level shifter 1V8/3V3
3	4	PA14	JTCK_SWCLK	R71 47R	T_SWCLKa
5	6	PB3	JTDO_SWO		level shifter 1V8/3V3
7	8	PA15	JTDI		
9	10	NRST	R69 47R		T_NRSTa
11	12	PA9	T_VCP_TX		level shifter 1V8/3V3
13	14				

Specific constraints for T_SWDIO and T_SWCLK (must be same length and must be Shielded).

SW LEVEL SHIFTER FOR MCU_1V8

MCU_1V8/3V3 — ST_LINK_3V3

VDD — C104 100nF — C105 100nF — 3V3_STLK

U30 — SN74LVC2T45DCUT
VccB VccA
B1 A1
B2 A2
DIR GND

JTMS_SWDIO — R72 47R — T_SWDIOa — T_SWDIO_IN
JTDO_SWO — R70 47R — T_SWOa

R113 0R DNF
R114 100R DNF

DIR : LOW : Direction is B to A

MCU_1V8/3V3 — ST_LINK_3V3

VDD — C94 100nF — C95 100nF — 3V3_STLK

U27 — SN74LVC2T45DCUT
VccB VccA
B1 A1
B2 A2
DIR GND

R110 100K — T_VCP_TX — R66 47R — STLK_VCP_RX

R103 0R DNF

DIR : LOW : Direction is B to A

5V ST-LINK PROTECTION

5V_USB_CHGR — C54 1uF — R80 1K — R82 100K — 5V_STLK

U20 TPS2051C
IN OUT
EN FAULT
GND

T_PWR_EN — R73 10K

LD3 RED

STLK_OVC

C61 1uF

ST-LINK POWER 3V3 / 150mA

5V_EXT — BAT60JFILM — D7
5V_UCPD — BAT60JFILM — D6
5V_DC — BAT60JFILM — D5
5V_USB_CHGR — BAT60JFILM — D4

U23 LD3985M33R
Vin Vout
INH GND BYPASS

3V3_STLK

C53 1uF — C57 100nF — C62 10nF — C58 1uF — C56 100nF

ST-LINK USB CONNECTOR

5V_USB_CHGR
Capacitor close to pin 1
C107 100pF — U34 — ESDA7P120-1U1M

Matched Net Lengths [Tolerance = 2.5mm]
Impedance Constraint [Min = 85.00 Max = 95.00]

CN26 STLK — USB_Micro-B receptacle — USB_uB_105017-0001

VBUS
DM — USB_DEV_HS_CN_N
DP — USB_DEV_HS_CN_P
ID
GND
Shield
Shield
Shield
Shield
EXP
EXP

USB — USB — U36 — ECMF02-2AMX6 — USB — USB

NC GND
D- D-
D+ D+

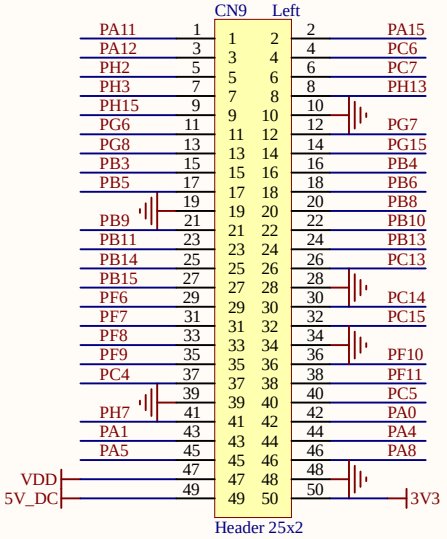
USB_DEV_HS_N
USB_DEV_HS_P

ADF1_CCK0 shared exclusif with TRACE-SWO and JTDO
UCPD_CC1 exclusive with JTAG_TDI

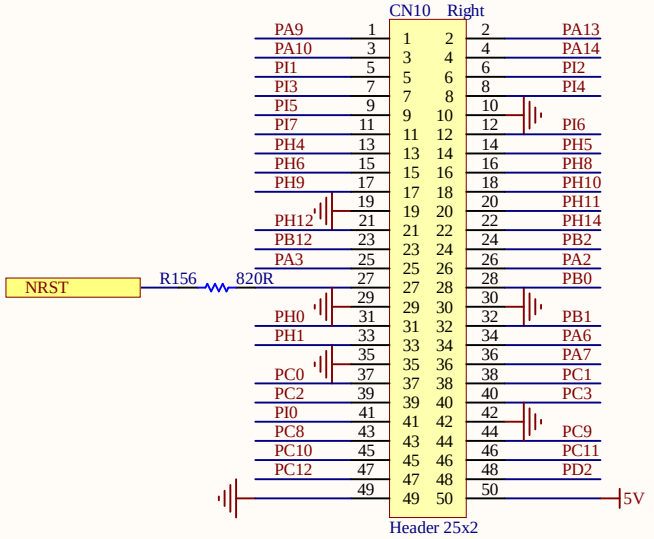
Title:	**STLINK V3E**
Project:	**STM32U575I-EV**
Variant:	U575AIQ
Revision:	C-02
Reference:	MB1550
Size:	A3
Date:	02 Sept 2021
Sheet:	23 of 24

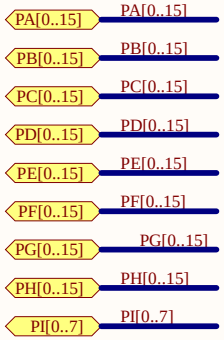
life.augmented

Placed close to MCU

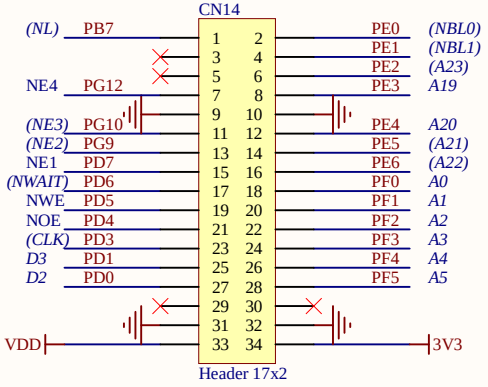


Placed close to MCU

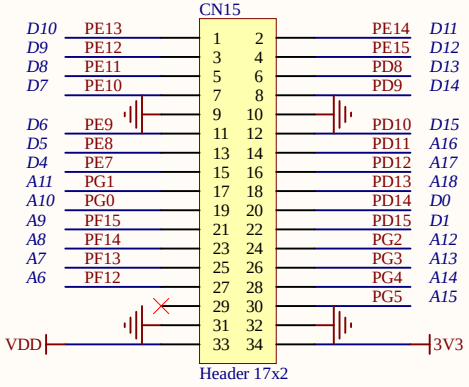




Placed close to memory

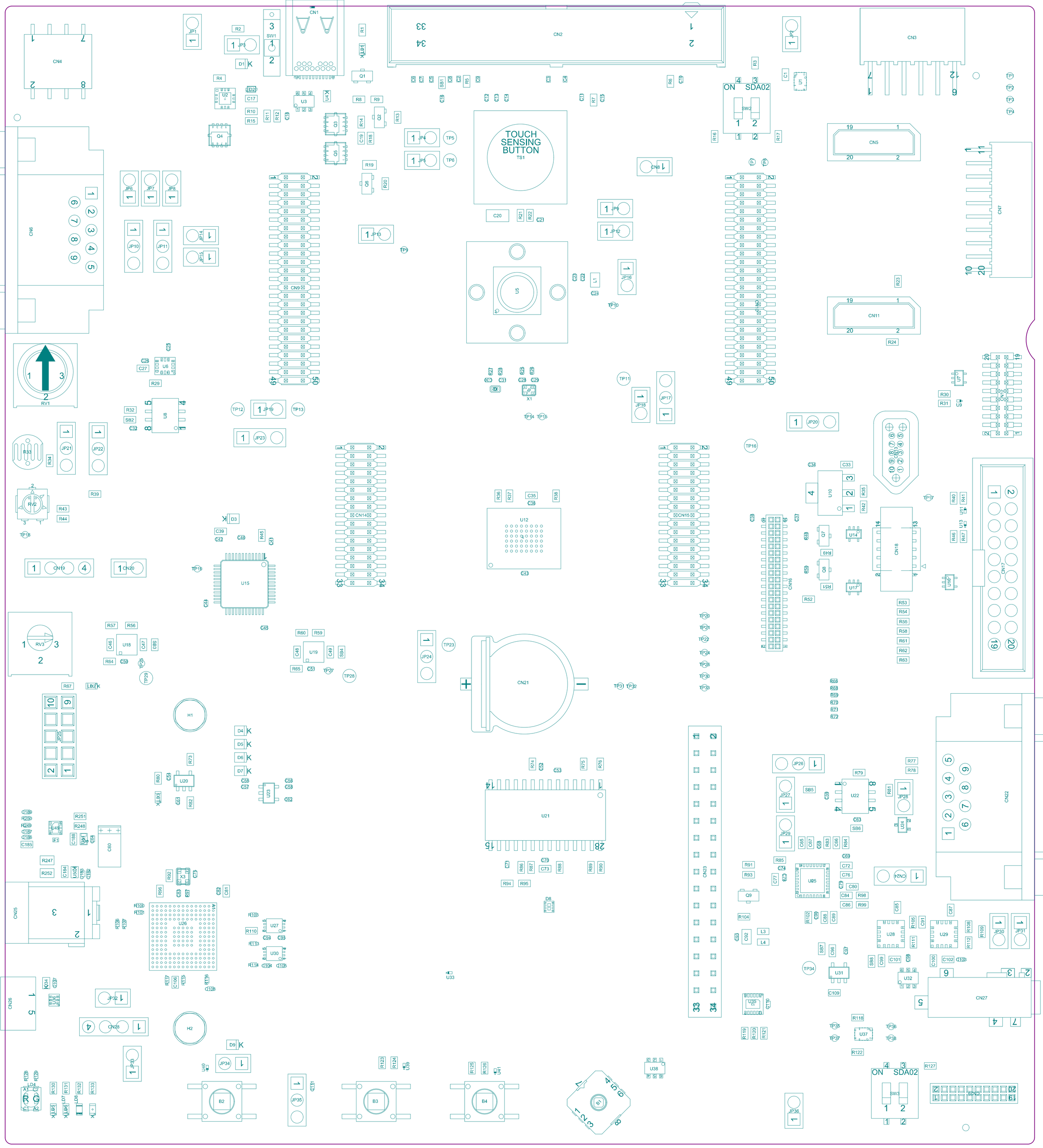


Placed close to memory



Title: **EXTENSION CONNECTORS**		
Project: **STM32U575I-EV**		
Variant: U575AIQ		
Revision: C-02		Reference: MB1550
Size: A4	Date: 02 Sept 2021	Sheet: 24 of 24

life.augmented



Project: STM32U575I-EV	
Layer: M14-Top Assembly	Gerber: .GM14
Variant: [No Variations]	Ref: MB1550
Date: 2020-NOV-13	Rev: C

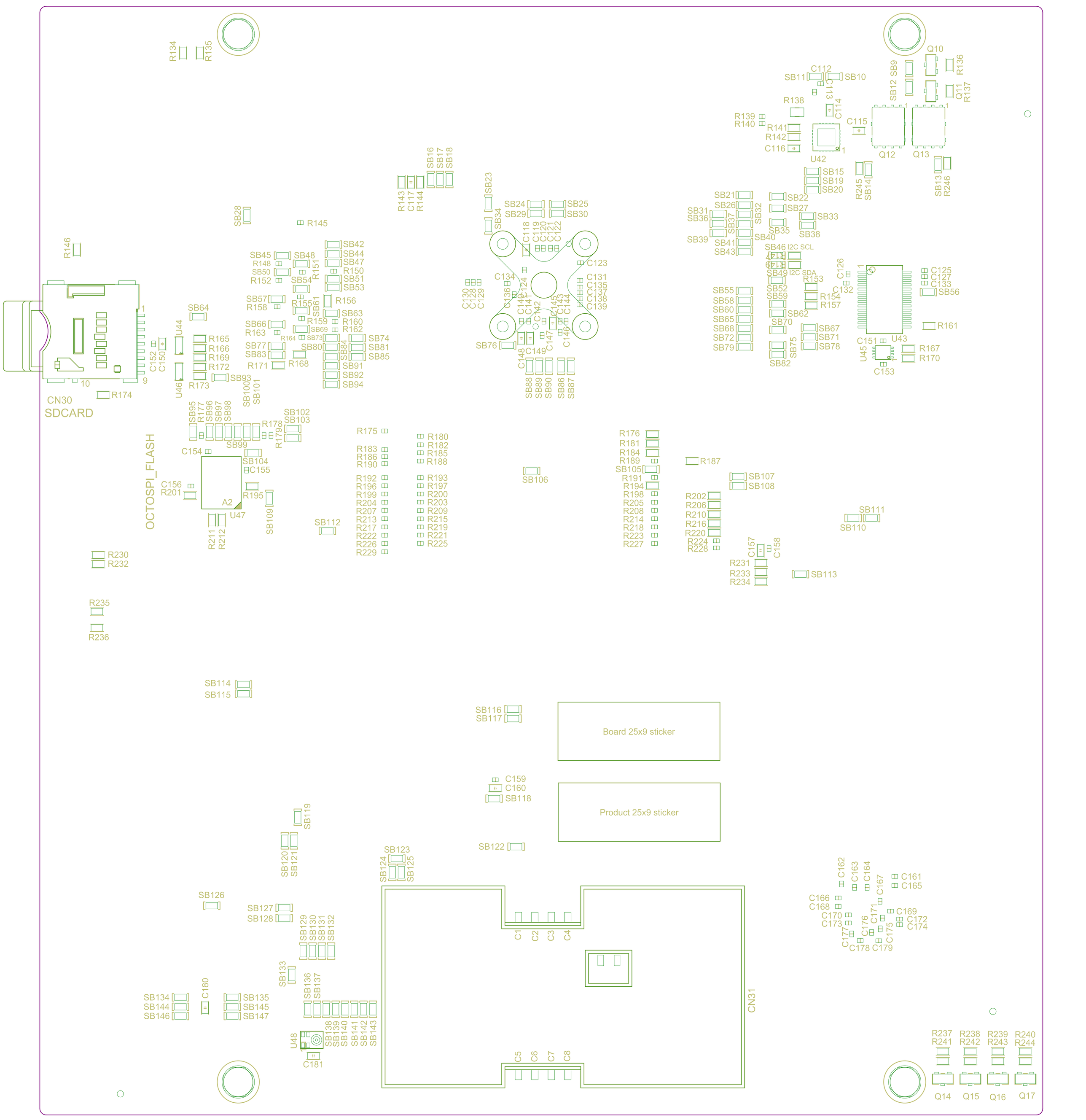
H3
H4
H5
H6
CN30
CN31
U42
U43
U44
U45
U46
U47
Q10
Q11
Q12
Q13
S1
S2
C1
C2
C3
C4
C5
C6
C7
C8
Board 25x9 sticker
Product 25x9 sticker

Project: STM32U575I-EV	
Layer: M15-Bottom Assembly	Gerber:.GM15
Variant: [No Variations]	Ref: MB1550
Date: 2020-NOV-13	Rev: C

MB1550C

MOTOR CONTROL
00 : SPI INTERFACE
10 : MikroBus INTERFACE
11 : UART INTERFACE

Top Overlay
Mid-1 Overlay
Mid-2 Overlay
Mid-3 Overlay
Bottom Overlay
Multi-Layer
.GTO
.GTP
.GM1
.GM2
.GM3
.GBO



M15-Bottom Assembly .GM15

Date: Sunday, ...